MultiQT notes

November 13, 2026

1 Notation

Unless stated otherwise, fix one disease, location and forecast horizon.

Forecasts. The index t labels a forecast round (reference week), and y_t is the outcome for its target week. Use s for an earlier round. At quantile levels $\mathcal{A} = \{\alpha_1 < \cdots < \alpha_m\} \subset (0,1)$, b_t^α is the base forecast and q_t^α the recalibrated forecast. The vectors $b_t, q_t \in \mathbb{R}^m$ collect all levels. Values are hospital counts unless a transformation is shown.

Coverage and loss. Write $\mathbf{1}\{\cdot\}$ for an indicator and $x_+ = \max(x, 0)$. Coverage at level α is $\mathrm{cov}_t^\alpha = \mathbf{1}\{y_t \le q_t^\alpha\}$, with target α. Quantile loss, with prediction z first and observation r second, is

$$\rho_\alpha(z, r) = \alpha(r - z)_+ + (1 - \alpha)(z - r)_+. \tag{1}$$

Ordering and states. Forecasts must be nondecreasing across levels. Write

$$\mathcal{K} = \{q \in \mathbb{R}^m : q^{\alpha_1} \le \cdots \le q^{\alpha_m}\},$$

and assume $b_t \in \mathcal{K}$. Projection onto a nonempty closed convex set C means the nearest point in Euclidean distance:

$$\Pi_C(x) = \arg\min_{z \in C} \|z - x\|^2.$$

A tilde marks hidden accumulated state; the untilded state is its projected version. Standard offsets use $\tilde{\theta}_t, \theta_t$; studentized offsets use $\tilde{u}_t, u_t$. Superscripts std, stu distinguish these methods, and η or η_t is the step size.

The methodological recursions forecast, observe y_t, then update. Delayed reports and revisions are introduced in Section 3.2; implementation, weighting and evaluation symbols are defined where they are used.

Pathwise view. No probability model is assumed for y_t, b_t or the reports. The recursions are deterministic functions of the data seen so far, and the statements below hold for each realized sequence. This is the setting of MultiQT [1] and of the quantile tracker [2], whose guarantees hold for any, even adversarial, sequence of outcomes. The guarantee is long-run empirical coverage: if $|y_t - b_t^\alpha| \le R$ for all t and α, then for every level

$$\Big|\frac{1}{T}\sum_{t=1}^{T} \mathrm{cov}_t^\alpha - \alpha\Big| \le \frac{2\|\tilde{\theta}_1\|}{\eta T} + \sqrt{\frac{m}{T} + \frac{2Rm^{3/2}}{\eta\, d_{\mathcal{A}} T}}, \qquad d_{\mathcal{A}} = \min_{\alpha \in \mathcal{A}} \min(\alpha, 1 - \alpha)$$

(their Theorem 1; their Proposition 7 adds a constant feedback delay). It says nothing about coverage in a given week, and with $m = 23$ levels and about 100 rounds per case the term $\sqrt{m/T} \approx 0.5$ alone allows a coverage gap of 0.5 at every level, whatever the rate: over our seasons

the bound does not constrain coverage. Coverage and quantile loss are therefore evaluated empirically, and differences between methods are those of one realized sequence. Section 17 checks how much such differences depend on the weeks and locations scored.

2 Methodological notes

2.1 Studentized score

Standard MultiQT tracks additive residual corrections. Studentization tracks scaled residuals, so changing the scale also changes the accumulated correction in outcome units. A falling scale can shrink a positive correction after a peak without erasing the hidden state. The quantile coverage target is unchanged; this construction alone does not establish calibration guarantees.

2.1.1 Standard MultiQT and score tracking

Starting with zero hidden state, MultiQT [1] projects the forecast but updates the unprojected state:

$$\theta_t = \Pi_{\mathcal{K}-b_t}(\tilde{\theta}_t), \qquad q_t^{\text{std}} = b_t + \theta_t, \tag{2}$$

$$\text{cov}_t^{\text{std},\alpha} = \mathbf{1}\{y_t \le q_t^{\text{std},\alpha}\}, \qquad \tilde{\theta}_{t+1}^{\alpha} = \tilde{\theta}_t^{\alpha} - \eta(\text{cov}_t^{\text{std},\alpha} - \alpha). \tag{3}$$

For a score $S_t(y)$, let $r_t = S_t(y_t)$ be its observed value and c_t its threshold; the target miscoverage is $1-\alpha$. The scalar score tracker of Angelopoulos et al. [2] uses

$$\mathcal{P}_t = \{y : S_t(y) \le c_t\}, \qquad \text{err}_t = \mathbf{1}\{r_t > c_t\}, \tag{4}$$

and updates

$$c_{t+1} = c_t + \eta(\text{err}_t - (1-\alpha)) = c_t - \eta(\mathbf{1}\{r_t \le c_t\} - \alpha). \tag{5}$$

This is a quantile-loss subgradient step. In the threshold argument,

$$\partial_z \rho_\alpha(z, r) = \begin{cases} \{-\alpha\}, & z < r, \\ [-\alpha, 1-\alpha], & z = r, \\ \{1-\alpha\}, & z > r, \end{cases} \tag{6}$$

and the selected subgradient, including ties, is

$$g_\alpha(z; r) = \mathbf{1}\{r \le z\} - \alpha \;\in\; \partial_z \rho_\alpha(z, r). \tag{7}$$

Loss depends on distance from the threshold; the update depends only on which side contains the observation.

2.1.2 Residual coordinates

The signed residual score recovers standard MultiQT:

$$S_t^{\text{std},\alpha}(y) = y - b_t^{\alpha}, \qquad r_t^{\text{std},\alpha} = y_t - b_t^{\alpha}. \tag{8}$$

Its threshold θ_t^{α} defines the forecast endpoint and coverage:

$$\{y : S_t^{\text{std},\alpha}(y) \le \theta_t^{\alpha}\} = (-\infty, b_t^{\alpha} + \theta_t^{\alpha}] = (-\infty, q_t^{\text{std},\alpha}], \tag{9}$$

$$\text{cov}_t^{\text{std},\alpha} = \mathbf{1}\{r_t^{\text{std},\alpha} \le \theta_t^{\alpha}\}. \tag{10}$$

Thus

$$\tilde{\theta}_{t+1}^{\alpha} = \tilde{\theta}_t^{\alpha} - \eta(\mathbf{1}\{r_t^{\text{std},\alpha} \le \theta_t^{\alpha}\} - \alpha), \tag{11}$$

with the same loss in residual coordinates:

$$\ell_t^{\text{std}}(\theta) = \sum_{\alpha \in \mathcal{A}} \rho_\alpha(b_t^{\alpha} + \theta^{\alpha}, y_t) = \sum_{\alpha \in \mathcal{A}} \rho_\alpha(\theta^{\alpha}, r_t^{\text{std},\alpha}). \tag{12}$$

For one level there is no ordering constraint, so this is scalar quantile tracking. For several levels, evaluate the subgradient at the projected state and apply it to the hidden state.

2.1.3 Studentized residuals and their objective

Choose finite, positive scales σ_t^α from information available at the forecast cutoff and hold them fixed when differentiating. Define

$$S_t^{\mathrm{stu},\alpha}(y) = \frac{y - b_t^\alpha}{\sigma_t^\alpha}, \qquad r_t^{\mathrm{stu},\alpha} = \frac{y_t - b_t^\alpha}{\sigma_t^\alpha}. \tag{13}$$

The score is increasing, so inversion preserves the coverage event:

$$q_t^{\mathrm{stu},\alpha} = b_t^\alpha + \sigma_t^\alpha u_t^\alpha, \qquad \mathrm{cov}_t^{\mathrm{stu},\alpha} = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\}. \tag{14}$$

Quantile loss of the standardized scores is a weighted outcome-unit loss:

$$\ell_t^{\mathrm{stu}}(u) = \sum_{\alpha\in\mathcal{A}} \rho_\alpha(u^\alpha, r_t^{\mathrm{stu},\alpha}) = \sum_{\alpha\in\mathcal{A}} \frac{\rho_\alpha(b_t^\alpha + \sigma_t^\alpha u^\alpha, y_t)}{\sigma_t^\alpha}. \tag{15}$$

Its selected subgradient is

$$g_t^{\mathrm{stu},\alpha}(u_t) = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\} - \alpha = \mathrm{cov}_t^{\mathrm{stu},\alpha} - \alpha. \tag{16}$$

Indeed, away from ties the inverse scale in the loss cancels the scale from differentiation of the forecast:

$$\frac{\partial}{\partial u^\alpha} \frac{\rho_\alpha(b_t^\alpha + \sigma_t^\alpha u^\alpha, y_t)}{\sigma_t^\alpha} = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u^\alpha\} - \alpha. \tag{17}$$

At ties use (7). Starting from $\tilde{u}_1 = 0$, the update is

$$\tilde{u}_{t+1}^\alpha = \tilde{u}_t^\alpha - \eta g_t^{\mathrm{stu},\alpha}(u_t). \tag{18}$$

The projected threshold determines coverage; only the hidden threshold accumulates increments. This is not gradient descent on the unweighted outcome-unit loss in u: that gradient has an additional factor σ_t^α. The coding section also tests median-centred forecasts and nonzero initial states.

2.1.4 Ordering and projection

Ordering always concerns the returned forecasts. For standard residuals,

$$C_t^{\mathrm{std}} = \mathcal{K} - b_t, \qquad b_t + \theta \in \mathcal{K} \iff \theta \in C_t^{\mathrm{std}}, \tag{19}$$

$$\theta^{\alpha_i} - \theta^{\alpha_{i+1}} \le b_t^{\alpha_{i+1}} - b_t^{\alpha_i}, \qquad i = 1, \dots, m-1. \tag{20}$$

For studentized residuals, collect the scales in $A_t = \mathrm{diag}(\sigma_t^{\alpha_1}, \dots, \sigma_t^{\alpha_m})$. Then

$$C_t^{\mathrm{stu}} = \{u : b_t + A_t u \in \mathcal{K}\} = A_t^{-1}(\mathcal{K} - b_t). \tag{21}$$

This is a nonempty closed convex polyhedron containing zero, generally not a cone. Unequal scales do not preserve $\mathcal{K}$, so it cannot generally be written as $\mathcal{K} - A_t^{-1} b_t$. The unique Euclidean projection gives

$$u_t = \Pi_{C_t^{\mathrm{stu}}}(\tilde{u}_t), \qquad q_t^{\mathrm{stu}} = b_t + A_t u_t. \tag{22}$$

Changing variables to $q = b_t + A_t u$ yields weighted isotonic regression:

$$q_t^{\mathrm{stu}} = \arg\min_{q\in\mathcal{K}} \sum_{\alpha\in\mathcal{A}} \frac{(q^\alpha - b_t^\alpha - \sigma_t^\alpha \tilde{u}_t^\alpha)^2}{(\sigma_t^\alpha)^2}, \tag{23}$$

For positive-definite W, write $\Pi_{\mathcal{K}}^W(v) = \arg\min_{q\in\mathcal{K}} (q - v)^\top W (q - v)$ for weighted projection. Then the equivalent projection identity is

$$b_t + A_t \Pi_{A_t^{-1}(\mathcal{K} - b_t)}(\tilde{u}_t) = \Pi_{\mathcal{K}}^{A_t^{-2}}(b_t + A_t \tilde{u}_t). \tag{24}$$

Weights $1/(\sigma_t^\alpha)^2$ follow from Euclidean projection in score coordinates; ordinary residual coordinates give equal weights.

2.1.5 Common and constant scales

For a scale σ_t common to all levels, $A_t = \sigma_t I$, where I is the identity matrix. Positive scalar multiplication preserves $\mathcal{K}$:

$$C_t^{\text{stu}} = \mathcal{K} - b_t/\sigma_t, \qquad q_t^{\text{stu}} = \Pi_{\mathcal{K}}(b_t + \sigma_t \tilde{u}_t). \tag{25}$$

The projection is ordinary isotonic regression. This is standard MultiQT on b_t/σ_t and y_t/σ_t, with the returned forecast multiplied by σ_t. For constant σ, it is also equivalent to standard MultiQT on the original inputs after matching initialization and converting state and rate:

$$\tilde{\theta}_t = \sigma \tilde{u}_t, \qquad \eta_{\text{std}} = \sigma \eta_{\text{stu}}. \tag{26}$$

For varying σ_t, changing the scale rescales the accumulated state; changing only the current rate does not reproduce that effect.

The equivalence carries the pathwise guarantee (Section 1) over to a common scale: Theorem 1 of [1] holds for b_t/σ_t and y_t/σ_t, with R a bound on $|y_t - b_t^\alpha|/\sigma_t$, which the lower clip of the scales at $.05S_0$ (Section 3.5.2) keeps finite. Rescaling leaves coverage unchanged, so the bound applies to q_t^{stu}.

to check:

Per-level scales change the constraint set. At a tie $q_t^{\alpha_i} = q_t^{\alpha_{i+1}}$ both levels share one coverage indicator c, and the negative gradient keeps their order for $c = 0$ and for $c = 1$ if and only if

$$\frac{\alpha_i}{\alpha_{i+1}} \le \frac{\sigma_t^{\alpha_{i+1}}}{\sigma_t^{\alpha_i}} \le \frac{1-\alpha_i}{1-\alpha_{i+1}}, \tag{27}$$

that is, $\alpha\sigma_t^\alpha$ is nondecreasing and $(1-\alpha)\sigma_t^\alpha$ nonincreasing in α. Ties occur both above and below y_t, and a larger tied block needs only its adjacent pairs, so (27) for every adjacent pair is equivalent to the inward-flow condition of their Lemma 2. Their Proposition 5 then applies as for a common scale: the gradient is still $\text{cov} - \alpha$, and the loss (15) is the MultiQT loss of the studentized residuals, restorative by their Lemma 1 with $R = \max_{t,\alpha} |r_t^{\text{stu},\alpha}|$. If the scales satisfy (27) at every round, the bound of Section 1 holds with this R; otherwise inward flow can fail. For the hub levels the band is $[.4, 1.015]$ from .01 to .025, $[.9, 1.1]$ from .45 to .5 and $[.985, 2.5]$ from .975 to .99: below the median the scales may fall steeply towards it but rise only slowly, and the reverse above it. U-shaped scales such as the spread scale of `studentized_spread` (Section 3.5.2) stay in the band in most weeks (Section 4.5.2).

2.1.6 Scores from transformations

A strictly increasing transform ψ_t, fixed at the forecast cutoff and the same at every level, gives the score and forecast

$$S_t^\alpha(y) = \psi_t(y) - \psi_t(b_t^\alpha), \qquad q_t^\alpha = \psi_t^{-1}(\psi_t(b_t^\alpha) + \theta_t^\alpha). \tag{28}$$

Since ψ_t is increasing, $\psi_t(b_t) + \theta \in \mathcal{K}$ exactly when the forecasts are ordered: the constraint set is $\mathcal{K} - \psi_t(b_t)$, a translate of $\mathcal{K}$ as in standard MultiQT. Inward flow holds without condition, and the bound of Section 1 applies to $\psi_t(y_t)$ and $\psi_t(b_t)$, even if ψ_t changes from week to week; a common scale is the affine case $\psi_t(y) = y/\sigma_t$. The `log1p` and square-root trackers are of this form.

To first order $S_t^\alpha(y) \approx \psi_t'(b_t^\alpha)(y - b_t^\alpha)$ (Section 3.5.2): the derivative version of (28) is the studentized score with the per-level scale $\sigma_t^\alpha = 1/\psi_t'(b_t^\alpha)$, which needs the band (27). For the Box-Cox family on counts,

$$\psi_\lambda(x) = \frac{(1+x)^\lambda - 1}{\lambda} \ (\lambda > 0), \qquad \psi_0(x) = \log(1+x),$$

with $\lambda = 1$ the identity and $\lambda = 0$ `log1p`, the derivative scale is $\sigma_t^\alpha = (1 + b_t^\alpha)^{1-\lambda}$. Taken at the forecast median instead, $\sigma_t = (1 + b_t^5)^{1-\lambda}$ is one scale for all levels and keeps the band. Per-level Box-Cox scales grow with the level and break its lower-tail bound in almost every week (Section 4.5.2): the exact transform keeps inward flow, its derivative version does not.

The transforms of Uniejewski [21] and the other scores tested, with their origins, the methods that use them and whether they keep inward flow, are collected in Section 3.5.2.

2.1.7 What the tracker needs from a score

The pathwise view of Section 1 needs no model. To see what a score must do for a small rate to suffice, this subsection alone treats the outcomes as random. Fix a level α and a score S_t^α of the form (13) or (28), and drop α from the notation: $r_t = S_t^\alpha(y_t)$ is the realized score, $\tilde{u}_t$ the hidden threshold, u_t its projected version and $\mathrm{cov}_t = \mathbf{1}\{r_t \le u_t\}$ the coverage, as in (14). Leave out the projection and the feedback delay, which change the round at which cov_t enters but not the argument, so that $u_t = \tilde{u}_t$. Let $\mathcal{I}_t$ be the information at the cutoff of round t (the reports and forecasts known then, so also b_t and σ_t), and

$$F_t(s) = P(r_t \le s \mid \mathcal{I}_t), \qquad u_t^\star = F_t^{-1}(\alpha),$$

the conditional distribution function of the score and its conditional α-quantile, with density $f_t = F_t'$. The update (18) is

$$\tilde{u}_{t+1} = \tilde{u}_t + \eta(\alpha - \mathrm{cov}_t), \qquad \mathbb{E}[\tilde{u}_{t+1} - \tilde{u}_t \mid \mathcal{I}_t] = \eta(\alpha - F_t(\tilde{u}_t)). \tag{29}$$

The mean step depends on F_t only through its value at $\tilde{u}_t$: it is zero at $u_t^\star$, positive below it and negative above it. The tracker is a stochastic approximation of the root $u_t^\star$ [3]: its step is the subgradient (7) of the quantile loss, whose expectation is minimized at the conditional quantile [4], and it uses nothing else of F_t. The same recursion is the adaptive specification of CAViaR [20]; adaptive conformal inference [6] applies it to the level of a conformal quantile.

Quantile stability. Call the score quantile-stable at level α if $u_t^\star = u^\star$ for every t. The mean, spread, shape and the other quantiles of F_t may change; only the α-quantile must stay put. Stationarity, the same F_t in every round, is far stronger. Under quantile stability every round has the same target. If $f_t(u^\star) \ge f > 0$ in every round, the mean step (29) points towards $u^\star$ from both sides: with decreasing steps ($\sum_t \eta_t = \infty$, $\sum_t \eta_t^2 < \infty$) $\tilde{u}_t$ converges to $u^\star$ [3, 5]; with a constant rate it settles near $u^\star$ and fluctuates around it. Coverage then holds round by round, not only on average: $P(\mathrm{cov}_t = 1 \mid \mathcal{I}_t) = F_t(\tilde{u}_t) \approx \alpha + f_t(u^\star)(\tilde{u}_t - u^\star)$.

Why the fluctuation is of order $\sqrt{\eta}$. Write $e_t = \tilde{u}_t - u^\star$ for the error of the threshold and $\xi_t = \mathrm{cov}_t - F_t(\tilde{u}_t)$ for the surprise in the coverage, a martingale difference with conditional variance $F_t(\tilde{u}_t)\{1 - F_t(\tilde{u}_t)\} \approx \alpha(1-\alpha)$. Near the root $F_t(\tilde{u}_t) \approx \alpha + f e_t$, and the update becomes

$$e_{t+1} = e_t - \eta(F_t(\tilde{u}_t) - \alpha) - \eta\xi_t \approx (1 - \eta f)\, e_t - \eta\xi_t, \tag{30}$$

an autoregression with coefficient $1 - \eta f$ and innovations of standard deviation $\eta\sqrt{\alpha(1-\alpha)}$. Its stationary variance is

$$\mathrm{Var}(e_t) \approx \frac{\eta^2\alpha(1-\alpha)}{1-(1-\eta f)^2} = \frac{\eta\,\alpha(1-\alpha)}{f\,(2-\eta f)} \approx \frac{\eta\,\alpha(1-\alpha)}{2f}. \tag{31}$$

Each step has size $\eta\alpha$ or $\eta(1-\alpha)$ and a random sign, and the pull towards $u^\star$ removes the share ηf of the error per round, so the threshold remembers about $1/(\eta f)$ rounds; a random walk of $1/(\eta f)$ steps of size η spreads by $\eta/\sqrt{\eta f} = \sqrt{\eta/f}$. In coverage the error is $f e_t$, with standard deviation $\sqrt{\eta f \alpha(1-\alpha)/2}$: calibrated on average, the coverage of a given round differs from α by an amount that shrinks only like $\sqrt{\eta}$. This is the stationary behaviour of constant-step stochastic approximation: the iterates form a Markov chain whose stationary distribution concentrates at the root with a spread of order $\sqrt{\eta}$ [8, 9].

A moving quantile. Without quantile stability the tracker chases a moving root. With $e_t = \tilde{u}_t - u_t^\star$ and the move of the root $\Delta_t = u_{t+1}^\star - u_t^\star$, (30) gives to first order

$$\mathbb{E}[e_{t+1} \mid \mathcal{I}_t] \approx (1 - \eta f)\, e_t - \Delta_t. \tag{32}$$

A root that moves by Δ per round leaves the threshold behind by $\Delta/(\eta f)$, a coverage error of about $|\Delta|/\eta$, while the noise (31) grows like $\sqrt{\eta}$. The mean squared error of the threshold, $\eta\alpha(1-\alpha)/(2f) + \Delta^2/(\eta f)^2$, is smallest at

$$\eta^\star = \Big(\frac{4\Delta^2}{f\,\alpha(1-\alpha)}\Big)^{1/3} :$$

the rate trades lag against noise, as in the rate grid of Section 4.2; the best rate grows with the drift and depends on the density f of the score, so on its scale. This lag–noise trade-off of constant steps is classical in adaptive filtering [10, 11]; for the step size of adaptive conformal inference Zaffran et al. [7] and Gibbs and Candès [22] combine several step sizes by expert weighting, as the rate-grid mixtures do here. A scale σ_t that keeps $u_t^\star$ constant sets $\Delta = 0$, removes the lag and lets the rate be small; this is what the choice of scale is for. A rate proportional to the spread of the recent errors, $\eta_t \propto 1/f$ like Delphi's adaptive rate (86), keeps both terms proportional to that spread: the same trade-off at every level of incidence, but no adaptation to the drift Δ.

What the drift ratio tests. Under quantile stability, the hits of the constant threshold, $\mathrm{cov}_t^\star = \mathbf{1}\{r_t \le u^\star\}$, satisfy $\mathbb{E}[\mathrm{cov}_t^\star \mid \mathcal{I}_t] = \alpha$: $\mathrm{cov}_t^\star - \alpha$ is a martingale difference sequence, uncorrelated over time with conditional variance $\alpha(1-\alpha)$, so its window averages vary as much as those of independent draws, or of the same hits in permuted order. The drift ratio of Section 4.5.2, with the hindsight α-quantile in place of $u^\star$, is then close to one. A moving root and misses that come in runs both break $\mathbb{E}[\mathrm{cov}_t^\star \mid \mathcal{I}_t] = \alpha$ and raise it. This is the conditional-coverage hypothesis of hit-sequence backtests (Christoffersen [19]; the dynamic quantile test of Engle and Manganelli [20]).

Why MultiQT is still needed. Quantile stability makes the target constant, not known. The base is calibrated at level α only if $u^\star = 0$ ($r_t \le 0$ exactly when $y_t \le b_t^\alpha$); its miscalibration is the unknown $u^\star$, which has to be learned from the reports as they arrive. Under quantile stability the natural estimator is the empirical α-quantile of all past scores, which the tracker approaches with decreasing steps; a static recalibration $q_t^\alpha = b_t^\alpha + \sigma_t \hat{u}^\star$, as in split conformal prediction with normalized scores [12], would then be enough. A constant rate gives up some of that efficiency in exchange for following the drift the scale leaves behind, and its guarantee (Section 1) needs no stability at all; the projection also keeps the levels ordered. A good scale thus turns the tracker's task from tracking a moving target into estimating a constant, and the tracker remains the safeguard where the scale falls short.

Relation to the literature. Each piece of this subsection is standard: the stochastic approximation of a quantile [3, 5], the $\sqrt{\eta}$ noise and the Δ/η lag of constant steps [10, 11, 8], and the hit-sequence tests [19, 20]. As far as we know, using quantile stability as the criterion for choosing the score of the tracker (Section 4.5.2) is the only addition.

2.2 The leak as a ridge penalty

Delphi's leak (`leaky`, (61)) shrinks the hidden state by the share δ before each update. For one level and one report per round, leaving out the projection,

$$\tilde{\theta}_{t+1} = (1-\delta)\tilde{\theta}_t - \eta g_t, \qquad g_t = \mathrm{cov}_t - \alpha. \tag{33}$$

This is exactly the subgradient step with step η on the quantile loss plus a ridge penalty:

$$\ell_t^\lambda(\theta) = \rho_\alpha(b_t + \theta, y_t) + \frac{\lambda}{2}\theta^2, \qquad \tilde\theta_t - \eta(g_t + \lambda\tilde\theta_t) = (1 - \eta\lambda)\tilde\theta_t - \eta g_t, \qquad \delta = \eta\lambda. \tag{34}$$

Over all levels the penalty is $\frac{\lambda}{2}\|\tilde\theta\|^2$, and the projection is unchanged; for a studentized score the same holds for $\tilde u_t$ in units of σ_t. The leak is thus weight decay, a Tikhonov penalty that pulls the correction towards zero, that is towards the base. In adaptive filtering the same device is the leaky LMS algorithm, which minimizes the mean squared error plus a ridge term [13]. Four consequences:

- **Bounded correction.** Since $|g_t| \le \max(\alpha, 1-\alpha)$, (33) started at zero keeps $|\tilde\theta_t| \le (\eta/\delta)\max(\alpha, 1-\alpha) = \max(\alpha, 1-\alpha)/\lambda$: with $\delta = .1$ and $\eta = \kappa S_0$ no level moves by more than $10\kappa S_0 \max(\alpha, 1-\alpha)$ (more only when several reports arrive in one round).
- **Shrinkage.** In expectation the leaky tracker settles where $\alpha - F_t(\theta) = \lambda\theta$ (F_t the conditional distribution of the residual $y_t - b_t^\alpha$, as in Section 2.1.7): not at the α-quantile $\theta^\star$ of the residual but between it and the base. With density f at $\theta^\star$, $\theta_\lambda \approx \theta^\star/(1 + \lambda/f)$: the leak keeps the share $1/(1 + \lambda/f)$ of the correction the base needs and trades calibration for stability.
- **Finite memory.** Unrolled from zero, $\tilde\theta_t = -\eta\sum_{s<t}(1-\delta)^{t-1-s} g_s$, an exponentially weighted sum of the past coverage errors with effective memory $1/\delta = 10$ rounds (half-life $\log 2/(-\log .9) \approx 6.6$ rounds). The weekly starts of `restart_recent`, which keep the last eight copies, are a sharp-cutoff version of the same idea.
- **Bayesian view.** With the asymmetric Laplace likelihood $p(y_t \mid \theta) \propto \exp\{-\rho_\alpha(b_t + \theta, y_t)/\tau\}$ of Bayesian quantile regression [14] and the prior $\theta \sim N(0, \tau/\lambda)$, ℓ_t^λ/τ is the negative log posterior of one round: the leaky tracker is an online stochastic-gradient estimate of the posterior mode, with the prior centred at the base, and λ is the prior precision relative to the likelihood scale τ.

Choosing the penalty. The leak has one parameter, $\lambda = \delta/\eta$; Delphi fixes $\delta = .1$. Three routes:

- **Criteria for the batch problem.** On a window of W rounds, the constant correction minimizing $\sum_s \rho_\alpha(b_s + \theta, y_s) + \frac{\lambda}{2}W\theta^2$ is a ridge-penalized quantile regression with an intercept only; λ could be chosen by the Schwarz criterion with the number of interpolated observations as degrees of freedom [15], by generalized approximate cross-validation [16], or through a prior on λ in the Bayesian model [17]. With one offset per level and 20 rounds these criteria have little to choose from (the effective degrees of freedom lie between 0 and 1); they would matter if the correction had more parameters, for example one per season phase.
- **Prequential choice**, the natural one online: replay the tracker over the last 20 rounds for a small grid of δ and keep the value whose issued forecasts had the smallest quantile loss, re-chosen every week [18]: a rolling tuning without a separate tuning season.
- **Mixtures**: run leaky trackers with several δ as experts and weight them by their recent losses, as the rate-grid mixtures do for η (Section 3.5.3).

With an adaptive rate η_t, a fixed δ means the penalty $\lambda_t = \delta/\eta_t$, which weakens when the errors grow; a fixed ridge penalty needs $\delta_t = \lambda\eta_t$ instead. REF-op (Section 3.5.4) penalizes only its fast offset; its slow offset is unpenalized, so the ridge decides how much of the correction the fast part may hold.

2.3 Revised reports as labels: corrected labels

The update (3) uses the outcome only through the coverage indicator. With delayed feedback (Section 3.2) that indicator is computed from the report current when row s is learned, which is revised later, while the forecasts are scored against the latest count, Y_s. The pathwise guarantee of Section 1 holds for the sequence the tracker learns from: at delay 0 the tracker is calibrated for first reports, not for Y_s. First reports are low (median 0.93 of the latest count, Section 4.3), so the tracker learns to forecast too low. Delay 1 improves the labels at the cost of one week; the weekly re-run improves those of older rows but learns the newest from its first report. This subsection keeps the timing of delay 0 and corrects the label instead: the indicator of a young report is replaced by the probability that the count lies at or below the forecast, given the reports.

Reports as measurements of one count. The reports of a target week are not independent draws around Y_s. They are nested (a later report adds late admissions to an earlier one) and almost always low: on the horizon-0 rows of the 10 states, 2–3% of first reports lie above the latest count, and the median ratio of first report to latest count is 0.91–0.94 up to peaks, after peaks and elsewhere, and 0.97–0.99 one week later. In the terms of the literature on data revisions [24], the reports are neither *noise* (count plus an independent error) nor *news* (the best estimate so far, with unpredictable revisions): the revisions are predictable, the right truncation that epidemic nowcasting models [25, 26]. A report of age a (days since the row's first report), $y_s^{(a)}$, is thus a measurement of Y_s whose error depends on a and can be learned from rows whose reports have matured.

Corrected labels. Let $\mathcal{I}_s$ be everything published by the cutoff at which row s is learned; q_s was issued earlier and is part of it. Replace the indicator by a probability $\hat{p}_s(z) \approx P(Y_s \le z \mid \mathcal{I}_s)$ at the issued forecast:

$$\tilde{\theta}^\alpha \leftarrow \tilde{\theta}^\alpha - \eta(\hat{p}_s(q_s^\alpha) - \alpha). \tag{35}$$

The test below estimates $\hat{p}_s$ from the location's own revisions: for the last K rows j whose first report is at least M weeks old at the cutoff ι, with $y_j^{(a)}$ their report at the age of $y_s^{(a)}$ and $y_j^{[\iota]}$ their current report,

$$d_j = \log \frac{1 + y_j^{[\iota]}}{1 + y_j^{(a)}}, \qquad \tilde{y}_{s,j} = (1 + y_s^{(a)})\, e^{d_j} - 1, \qquad \hat{p}_s(z) = \frac{1}{K} \sum_j \mathbf{1}\{\tilde{y}_{s,j} \le z\}, \tag{36}$$

with $K = 26$ and $M = 2$; a report at least M weeks old, or one with fewer than 8 matured rows behind it, keeps its indicator. The plausible counts $\tilde{y}_{s,j}$ are samples of the unknown count, drawn from the current report and the past revisions rather than from the successive reports themselves. Three properties make (35) an ordinary MultiQT step in disguise: it is the average of K standard steps, one per plausible count, each with rate η/K; $\hat{p}_s$ is nondecreasing in z, so the levels stay ordered; and $\hat{p}_s$ is 1 above every plausible count and 0 below, so the state is pushed back as with bounded outcomes. Since first reports are lower bounds, a first report above q_s^α gives $\hat{p}_s(q_s^\alpha) = 0$ in practice: the miss is known at once, as for a right-censored observation in censored quantile regression [27], and only "covered" is uncertain. A nowcasting model [25, 26] could replace (36).

Later reports. Two ways bring in the reports that arrive later. (i) *Corrected now, replaced at delay d* (stateful): learn (35) at $\tau_0(s)$ and, at $\tau_d(s)$, take the step

$$\tilde{\theta}^\alpha \leftarrow \tilde{\theta}^\alpha - \eta\Big(\mathbf{1}\{y_s^{[\iota_{\tau_d(s)}]} \le q_s^\alpha\} - \hat{p}_s(q_s^\alpha)\Big). \tag{37}$$

Since q_s is the issued forecast, the two steps add up to the delay-d step (42): the corrected label moves part of it forward. This is the reduction of delayed to optimistic online learning [29], with the expected step of a row whose feedback is outstanding as the hint. (ii) *Re-run with corrected labels*: in (45), rows whose current report is younger than M weeks take $\hat{p}_s$ from that report instead of the indicator.

Proposition 2.1 (Coverage with corrected labels)**.** *Fix a level α and number the learned rows $s = 1, \dots, n$ in the order they are learned. Row s is learned once, with a label $\ell_s \in [0,1]$ (a corrected label $\hat{p}_s(q_s^\alpha)$ or an indicator), and $C_s = \mathbf{1}\{Y_s \le q_s^\alpha\}$ is its coverage of the count it is scored against. If the hidden state stays bounded, $|\tilde{\theta}_t^\alpha| \le B$, then*

$$\frac{1}{n}\sum_{s=1}^{n}(C_s - \alpha) = \underbrace{\frac{1}{n}\sum_{s=1}^{n}(\ell_s - \alpha)}_{\text{tracker term}} + \underbrace{\frac{1}{n}\sum_{s=1}^{n}(C_s - \ell_s)}_{\text{label error } \bar{e}_n}, \tag{38}$$

and:

(a) the tracker term is at most $2B/(\eta n)$ in absolute value, for every sequence of labels and counts;

(b) if the labels are calibrated, $E[C_s \mid \mathcal{I}_s] = \ell_s$, then $E[\bar{e}_n] = 0$; if moreover Y_s is known by the time row $s + L$ is learned (C_s is $\mathcal{I}_{s+L}$-measurable), then $\mathrm{Var}(\bar{e}_n) \le (2L+1)/(4n)$, so the coverage of the latest counts approaches α up to a label error of order $\sqrt{L/n}$;

(c) with the indicator of the first report, $\ell_s = \mathbf{1}\{y_s^{(0)} \le q_s^\alpha\}$,

$$\bar{e}_n = \frac{1}{n}\#\{s : Y_s \le q_s^\alpha < y_s^{(0)}\} - \frac{1}{n}\#\{s : y_s^{(0)} \le q_s^\alpha < Y_s\},$$

so with low first reports the coverage of the latest counts falls short of α by the share of rows whose quantile lies between the first report and the count;

(d) with the replacement (i), the rows replaced so far carry the net label $\mathbf{1}\{y_s^{[\iota_{\tau_d(s)}]} \le q_s^\alpha\}$, and their tracker term is at most $(2B/\eta + P)/n$ with P rows still pending: the long-run coverage is that of delay d, with the label error of the delay-d report.

Proof. The hidden state is not projected, so learning row s moves it by $-\eta(\ell_s - \alpha)$, and $\tilde{\theta}_{\mathrm{end}}^\alpha - \tilde{\theta}_1^\alpha = -\eta\sum_s(\ell_s - \alpha)$; with $|\tilde{\theta}^\alpha| \le B$ this gives (a). For (b), ℓ_s is $\mathcal{I}_s$-measurable, so $E[C_s - \ell_s] = E\big[E[C_s \mid \mathcal{I}_s] - \ell_s\big] = 0$. Let $\xi_s = C_s - \ell_s$; then $E[\xi_s \mid \mathcal{I}_s] = 0$ and $\mathrm{Var}(\xi_s) = E[\ell_s(1-\ell_s)] \le 1/4$. For $s' > s + L$, ξ_s is $\mathcal{I}_{s'}$-measurable, so $E[\xi_s\xi_{s'}] = E\big[\xi_s\, E[\xi_{s'} \mid \mathcal{I}_{s'}]\big] = 0$. At most $n(2L+1)$ ordered pairs have $|s - s'| \le L$, each with $|E[\xi_s\xi_{s'}]| \le 1/4$ by Cauchy–Schwarz, so $\mathrm{Var}(\sum_s \xi_s) \le n(2L+1)/4$. In (c), $C_s - \ell_s$ is 1 when $Y_s \le q_s^\alpha < y_s^{(0)}$, -1 when $y_s^{(0)} \le q_s^\alpha < Y_s$ and 0 otherwise. In (d), the two steps of a replaced row add up to $-\eta(\mathbf{1}\{y_s^{[\iota_{\tau_d(s)}]} \le q_s^\alpha\} - \alpha)$, and each pending row contributes at most 1 in absolute value. □

Remark. The bound B is where the counts enter. For one level without projection (the quantile tracker of [2]), if every plausible count lies within R of the base, an offset above R takes only downward steps, except for rows issued before it got there, so the state stays within R plus the steps of the rows still pending. For MultiQT, Theorem 1 of [1] bounds the hidden state for outcomes within R of the base; a corrected step is an average of such steps, and we expect their argument to carry over, but we have not re-derived its constants. Part (a) is pathwise, like Section 1. Part (b) is the only probabilistic statement, and it concerns the revision model, not the counts: a miscalibrated $\hat{p}$ shifts the long-run coverage by its average calibration error $\frac{1}{n}\sum_s(E[C_s \mid \mathcal{I}_s] - \ell_s)$, which can be checked on past rows (below).

Where the pieces come from.

- **Unbiased steps.** The decomposition (38) into a pathwise tracker term and a martingale-difference label error is the standard analysis of stochastic approximation with noisy, unbiased steps [3, 8]. Condition (b) asks that the corrected step be an unbiased estimate, given $\mathcal{I}_s$, of the step with Y_s; replacing the indicator by its conditional expectation also lowers the variance of the step (Rao–Blackwell).
- **Delayed feedback.** In online learning the regret grows with the delay [28]; optimistic hints for the rounds whose feedback is outstanding recover part of it [29], there for subseasonal climate forecasts. (i) is that reduction with the corrected step as the hint.
- **Corrupted feedback.** Online conformal prediction whose coverage indicators are flipped by an unknown process keeps a long-run guarantee by compensating the expected flips [30]. A low first report flips "missed" to "covered" whenever the quantile lies between report and count, as in (c).
- **Labels that arrive late.** In online advertising a conversion can follow a click by weeks; the label is modelled jointly with its delay rather than taken at face value [31].
- **Lower bounds and nowcasts.** Censored quantile regression uses lower bounds as observations [27]; nowcasting models give the distribution of the count from incomplete reports [25, 26].
- **Delphi.** Delphi's calibration notebooks compare learning from finalized values, from the value at the reveal round (stateful) and the weekly re-run (E00), extra delays of 0 to 3 weeks at each rate (E09), and the gap between the learned and the final value (E21); production re-runs with one extra week. None corrects the learned value; Delphi's NHSN nowcasts (an evaluation of the `baselinenowcast` package) and revision-aware forecasters are separate from the tracker [23].

We have not found the combination, a nowcast of the count as the tracker's label with coverage of the latest counts, in this literature or in Delphi's code; each piece is standard.

First test. Fixed MultiQT at equal rates in the three designs, the corrected variants and a hindsight reference that learns the latest count at the timing of delay 0; the 10 states, horizons -1 to 3, evaluation period, scored against the latest counts, cases weighted by their weeks as in script 02. Table 1 gives the overall change of loss and calibration error in % of the base's (negative is better).

Table 1: Corrected labels: overall loss and calibration error (CE) change versus the base (%), Fixed MultiQT at equal rates κ.

	COVID				Flu			
	$\kappa = .01$		$\kappa = .03$		$\kappa = .01$		$\kappa = .03$	
Design	loss	CE	loss	CE	loss	CE	loss	CE
delay 0	+4.0	+15	+9.7	−3	+0.9	−45	+6.7	−33
delay 1	+3.4	−25	+11.5	−38	+1.2	−55	+8.4	−46
re-run	+3.0	−38	+9.7	−46	+1.1	−51	+7.1	−53
delay 0, corrected labels	+2.5	−32	+8.8	−52	+1.0	−47	+6.8	−56
(i) corrected, replaced at delay 1	+2.8	−26	+9.5	−40	+0.9	−57	+6.9	−52
(ii) re-run, corrected labels	+2.9	−39	+9.2	−55	+1.1	−47	+7.1	−54
delay 0, latest counts (hindsight)	+3.2	−36	+10.5	−54	+1.2	−45	+7.1	−60

- **The corrected labels are calibrated.** At the base quantiles of the evaluation rows, the label error $\bar{e}_n$, averaged in absolute value over the levels, is .009 (COVID) and .007 (flu), against .051 and .045 for the first report and .022 and .024 for the report one week later.
- **Learned at delay 0, they calibrate like the re-run or better and lose least overall.** For COVID at $\kappa = .03$: CE -52% against -3% (delay 0), -38% (delay 1) and -46% (re-run), loss $+8.8\%$ against $+9.7\%$, $+11.5\%$ and $+9.7\%$. Over $\kappa = .005, .01, .03, .1$ and both diseases they beat delay 1 on CE in 6 of 8 cells (not flu at $\kappa \leq .01$) and on loss in 7 of 8, and come close to the hindsight reference.
- **(i) calibrates like delay 1**, as Proposition 2.1(d) says: its long-run label is the report one week later, still 1–3% low. It loses less than delay 1 (COVID at $\kappa = .03$: $+9.5\%$ against $+11.5\%$).
- **(ii) improves the re-run a little**: COVID CE -39% and -55% against -38% and -46%; flu only from $\kappa = .03$ on.
- **After peaks the corrected labels lose what delay 0 gains there.** Delay 0's low labels push its forecasts down, which helps when the base is too high; with corrected labels the after-peak scores are those of the re-run (COVID at $\kappa = .01$: $+7.1\%$ loss against $+4.7\%$ at delay 0 and $+8.1\%$ in the re-run; hindsight reference $+8.9\%$). The late reaction after peaks is the dead time of $h + 1$ weeks between the newest reported week and the target, which no label removes.

Plots and scripts. `agent/outputs/soft_labels/soft_labels_tradeoff.pdf`: loss change against CE along the rates, per disease and phase; `agent/outputs/soft_labels/soft_labels_examples.pdf`: four example cases (COVID California and Texas, flu California and New York, $h = 2$). `agent/scripts/soft_labels/run.R` (with checks against `multiqt()` and the re-run of R/03), `agent/scripts/lib/soft_labels.R`. Not in R/; open choices are K, M, pooling the revisions over states and a nowcasting model for $\hat{p}$.

2.4 Questions and ideas

Which property of the base forecast should another score recalibrate? Increasing, invertible scores preserve the quantile coverage event; changing the target to PIT or highest-density-region calibration would require a separate formulation.

Open issue: rate tuning (to do). The rates R/05 tunes on the tuning season (late November 2024 to May 2025, one winter) are too large for the evaluation period: tuned once on the evaluation weeks, the same rule picks a smaller rate in 37 of 44 method–cell pairs, and the tuned rate costs 5 to 18 points of loss plus calibration change (Section 4.5.5). With the rate in hindsight, Fixed MultiQT beats every scale, so comparisons of methods at tuned rates partly compare how well each method tolerates a wrong rate. To do: tune on a longer history (the `windowed_seasonal` base starts in October 2023), choose rates on rolling windows or with a preference for small rates, compare with the untuned adaptive rates (Section 3.5.4), and recheck the comparisons of methods once the tuning is settled.

Towards untuned rates. The results support dropping the tuning, but not for Delphi's adaptive rate as such. For dropping it: one tuning season asks for rates that are too large (above); the elbow rule is unstable, pooling 10 states instead of 52 locations moved Fixed MultiQT's COVID rate at delay 0 from .01 to .1, and several methods sit at the edge of their grids (Table 7); a small rate does better without tuning: Fixed MultiQT at $\kappa = .01$, the low end of the grid of script 02, beats the tuned Fixed MultiQT on both loss and calibration error in every flu design and loses 15 to 42 points less for COVID, at a smaller calibration gain (Tables 6 and 8). Delphi's

adaptive rate is not what helps: it is the fixed multiplier .03 on a rolling S_0, the 90% quantile of the last 20 absolute errors, so it adapts to the size of the errors, not to their drift (Section 2.1.7). On counts it raises the rate at the peaks, where the errors are largest, and `fixed_adaptive` is beaten on both loss and calibration error by Fixed MultiQT at $\kappa = .01$ in all six cells (COVID at delay 0: +9.1% loss and +60% CE against +4.0% and +16%); on square-root counts, too, Delphi's constant rate (`sqrt_constant`) loses less than the adaptive one (`sqrt_adaptive`) in every cell. For a studentized tracker, whose score is already in units of its scale, the rolling quantile of the scores is nearly constant, and the adaptive rate is close to a fixed rate in units of σ_t. To do: replace R/05 by small fixed multipliers chosen once, not tuned (for example $\kappa = .01$ on counts and in units of the scaled methods' scales, Delphi's constants for its trackers), keep the replay of the first 20 weeks for S_0, choose the leak prequentially or by a mixture (Section 2.2), and rerun.

Idea: scores with a seasonal centre. No positive scale can remove the drift of the scores (Section 4.5.2): the base under-forecasts on the rise and over-forecasts after the peak, so the α-quantile of the residual changes sign at the peak, and a scale only rescales the residual. A seasonal scale (for example from the same weeks of earlier seasons) has the same limit. What could help is a seasonal centre: a score $(y - b_t^\alpha - \mu_t^\alpha)/\sigma_t$ with an expected bias μ_t^α for the phase of the wave, learned on earlier seasons. It needs the phase at the cutoff, which is known only in hindsight: real-time proxies (the base's own trend, the growth of the reports, the forecast-path indicators of R/02) are least reliable at the peak, where the sign changes, and one or two seasons are little to learn a bias profile from. The removed Decline MultiQT, which learned only in declining context, was a step in this direction.

Open issue: the exploding t-PIT scale (to do). The t-PIT scale (68) divides by the kernel density of the recent reports at the forecast median, which is almost zero when the median lies outside their range (late summer 2026; California COVID at delay 1 in March 2025). The guard against the running median does not contain it: the median includes the current week and is inflated when the explosion comes early, as in the S_0 rows of California, where every rate then fails and the tuning picks the edge of its grid (Section 4.8). To do: guard against the median of the past weeks only, after a minimum number of weeks; bound the normalized scale, e.g. to $[.05, 20]\, S_0$ or tighter; floor the density (a bandwidth tied to the forecast's spread, or the power scale where the median leaves the range of the reports); then rerun the t-PIT methods and their tuning alone.

3 Coding Updates

The tracker transform ψ is the identity, $\psi(x) = x$ (`identity`), for every method except `fixed_log` and `leaky_log`, which run standard MultiQT on $\psi(x) = \log(1 + x)$ (`log1p`). The transformed base is $\beta_t^\alpha = \psi(b_t^\alpha)$. Offsets θ_t are in these tracker units. The count divisor c_{scale} used for scoring is population/100,000 by default.

3.1 Data

Script 00 downloads CMU–TimeSeries hub forecasts and dated NHSN reports, and generates our own base forecasts where the required inputs are available. Script 01 prepares cases (one disease, location and forecast horizon) from the selected base forecasts and reports. Cases retain generation cutoffs, revisions, feedback schedules for delays 0–4, populations and filled-forecast flags.

Reference weeks are Saturdays (`reference_date`); the target week of y_t is `target_end_date` in forecasts and `target_date` in prepared cases. The information cutoff ι_t is normally the Wednesday of the reference week, or the actual generation date where known. Write $y_s^{[v]}$ for the latest report for the target week of row s, dated on or before cutoff v (archive field `version`). A valid report is finite, nonnegative and not excluded by review. "Latest counts" means the latest reports in the downloaded archive, through 30 October 2026.

Hub forecasts: the active design. The active design uses submitted CMU–TimeSeries forecasts for both diseases from reference date 23 November 2024. The stored quantiles are used subject to the cleaning and filling below. Feedback uses the report available at the relevant cutoff, while evaluation uses latest counts.

CMU–TimeSeries is an ensemble of Delphi models. Autoregressive quantile models use recent hospital admissions, sometimes with emergency-department data; linear-trend models extrapolate recent changes, and climatological models use patterns from earlier years. The ensemble takes a weighted average of the component forecasts at each quantile level. Components and weights can vary by date, location and horizon. We use these archived ensemble quantiles as b_t^α and apply our recalibration to them. See the CMU model description and the pinned ensemble code.

Target dates determine the split: tuning through 31 May 2025, then evaluation from June 2025 through September 2026. Overall scores use all available evaluation weeks; before/after-peak scores restrict these to the corresponding peak windows. Flu has long summer gaps in submitted forecasts, rather than an explicit winter-only evaluation rule. The flu winter 2024/25 peak belongs to tuning. Plotted target weeks after September 2026 are outside the evaluation period and are not scored.

Own `windowed_seasonal` forecasts: the longer-history alternative. The longer-history setup starts on reference date 14 October 2023, with tuning target dates through 27 April 2024 and evaluation after the reporting gap below, through 30 September 2026. Script 00 implements generation of both COVID and flu forecasts with unchanged, pinned Delphi code and packages and historical HHS reports for the earlier period.

`windowed_seasonal` is one autoregressive model, refitted at each reference date for each horizon. It fits quantile regression (solver `fn`, as in Delphi's production) to pooled training rows, using the latest available weekly admission count and its one-week lag. Counts undergo a fourth-root transform, centering and scaling within each location and source; predictions are transformed back to nonnegative counts and their quantiles sorted. For flu, the training window selects weeks near the corresponding time of year across available years. The COVID configuration uses no seasonal restriction despite the model name. These forecasts then serve as

the base for our recalibration. Script 00 generates horizons $0, \ldots, 3$ only. At $h = -1$ the target week is usually already reported, so the outcome equals the latest-count predictor: the fit is exact and all quantiles coincide (zero width). Delphi's submissions also take $h = -1$ from other models. Without it, the nowcast and revision indicators (Section **??**) are unavailable. See the pinned model implementation and COVID/flu settings.

Inputs for own flu forecasts. Flu adds historical ILI+ and adjusted FluSurv-NET training rows to the hospital counts; COVID uses hospital counts alone. The eight required CSVs have been downloaded in `aux_data.zip`: `flu_burden.csv`, `us_pop.csv` and six WHO/NREVSS laboratory files. Script 00 now extracts them to `external/delphi/aux_data/flusion_data/` and supplies them to the Delphi functions before forecast generation.

How earlier reports are reconstructed. Our NHSN archive contains historical counts, but report versions start on 19 November 2024. Script 00 now reconstructs earlier weekly HHS versions with the pinned Delphi aggregation script. For a Saturday target date, it sums admission days from the preceding Saturday through Friday; a weekly count is available only when all seven daily counts are present. The HHS versions precede the first NHSN version, which supersedes them. Base forecasts use reports available at the generation date. Script 01 uses HHS versions for earlier feedback where available, including later revisions in the re-run design; scoring uses latest NHSN counts.

Other limitations of the longer-history experiment. These are not flu-specific. Target weeks 4 May–2 November 2024 have incomplete reporting and are excluded from learning and scoring; our forecast schedule pauses across this gap. Stateful methods carry accumulated corrections and weights forward when rounds resume, rather than restarting. This reporting gap and differences between the earlier HHS and later NHSN sources affect both diseases. With four horizons instead of five, results pooled over horizons are not directly comparable with those of the hub forecasts.

Cleaning and filling.

1. **Missing forecasts.** Fill gaps of at most six rounds from the most recent earlier forecast for the same target: horizon h at reference date r uses $h + k$ at $r - 7k$ days, with the smallest available $k > 0$. Horizon 3 has no source. Filled rows are flagged and scored; unfillable rounds remain absent.
2. **Broken submissions.** Remove the known Delphi bugs: COVID 23 November 2024; COVID $h = -1$ on 25 April 2026; US flu 14 and 21 December 2024 and 4 January 2025. Also remove all-zero submissions across every location and quantile of a disease–round–horizon (flu $h = -1$, 22 November 2025).
3. **Missing reports.** If unpublished or currently invalid, stateful feedback waits and re-runs omit it; neither substitutes an older valid report. Keep the forecast timeline when latest truth is unavailable, but omit that target from scoring; earlier valid feedback can still be used. Targets 4 May–2 November 2024 are excluded from learning and scoring.
4. **Report errors.** A reviewed exclusion table masks identified erroneous report versions, retaining later corrections and their target weeks. Each exclusion records its evidence; decisions based on later revisions are retrospective cleaning. For example, Pennsylvania COVID for 16 November 2024 changed from 6,124 to 1,005 the next day.

Disease	Missing reference weeks	Unfilled target weeks, $h = 2$
COVID	25 January–1 February 2025	15 February 2025
Both	25 October–15 November 2025	15–29 November 2025
Flu	25 January 2025; 18 April 2026	None
Flu	7 June–11 October 2025	21 June–25 October 2025
Flu	6 June–3 October 2026	20 June–17 October 2026

The COVID $h = 2$ weight plot (`multiqt_weight_timelines/covid_horizon2_delay0.pdf`, p. 1) shows the first two gaps; only the first missing round can use an earlier $h = 3$ forecast. US flu also lacks target 4 January 2025 at $h = 2$ after successive broken submissions. Lighter bars denote filled rows. Lines break at gaps longer than 28 days and can bridge shorter missing intervals. Updates occur only at retained rounds: pending feedback waits across unfilled rounds, without resetting the state.

Reporting pause in autumn 2025. NHSN published no report between 26 September and 14 November 2025, a pause that coincides with the US federal government shutdown (1 October–12 November). Target weeks 27 September–8 November were first reported together on 14 November, 6 to 48 days after the week ended; weekly reports resumed on 19 November. Both diseases are affected. CMU kept submitting through 18 October and then skipped four rounds. Each round of 2025 around the pause, with its cutoff on the Wednesday before:

Round	Newest NHSN report at the cutoff	CMU forecast	Learned at delay 0
27 Sep	24 Sep, through week 20 Sep	submitted	week 20 Sep
4 Oct	26 Sep (revision)	submitted, no new counts	—
11 Oct	26 Sep	submitted, no new counts	—
18 Oct	26 Sep	submitted, no new counts	—
25 Oct	26 Sep	missing; filled for $h \leq 2$	—
1 Nov	26 Sep	missing; filled for $h \leq 1$	—
8 Nov	26 Sep	missing; filled for $h \leq 0$	—
15 Nov	26 Sep	missing; filled for $h = -1$	—
22 Nov	19 Nov, through week 15 Nov	submitted	weeks 27 Sep–15 Nov
29 Nov	26 Nov, through week 22 Nov	submitted	week 22 Nov

The trackers learn nothing for seven rounds: at delay 0 the last update before the pause is in the round of 27 September and the next in that of 22 November (delay 1: 4 October and 29 November; the re-run learns from the same reports). The forecasts of 4–25 October carry the corrections of late September, and the weeks of the pause are then learned in one step. CMU's forecasts of 4–18 October had no new NHSN counts, so they extrapolate one to three weeks further than their horizon says. The pause came right after the summer COVID wave: 45 of the 52 locations without the US have a COVID peak in August–September 2025 (the US on 6 September, 11,048 admissions, falling to 3,649 by 25 October), and 287 of the 800 COVID after-peak weeks (36%) are among the weeks first reported on 14 November. The COVID results after peaks thus partly measure trackers that could not see the decline. No flu peak is near the pause.

Locations. Every disease–location–horizon case learns independently. Detailed plots show the US, California, Texas, New York and Pennsylvania. Scores cover all 53 locations. Since 12 November 2026, tuning and every pooled summary, plot and check use the 10 largest states without Florida: California, Texas, New York, Pennsylvania, Illinois, Ohio, Georgia, North Carolina, Michigan and New Jersey (`pooled_locations`); before, they used the 52 non-US locations. The US is never pooled and is evaluated separately with the selected parameters. Forecasts are never averaged across locations.

3.2 Feedback

Feedback uses the latest report at each cutoff as defined in Section 3.1, independently of the recalibration method. Retrospective initialization is described in Section 3.3.

Stateful learning. Let ν_s be the publication date of the first valid report for row s. Wait d calendar weeks, giving maturity date $m_s^{(d)}$, then consume the newest report when valid at the next retained forecast cutoff with $t > s$. Write $\tau_d(s)$ for that round and $\mathcal{R}_t^{(d)}$ for the rows consumed before forecasting at t:

$$m_s^{(d)} = \nu_s + 7d \text{ days}, \tag{39}$$

$$\tau_d(s) = \min\{t > s : \iota_t \geq m_s^{(d)},\ y_s^{[\iota_t]} \text{ valid}\}, \qquad \mathcal{R}_t^{(d)} = \{s : \tau_d(s) = t\}. \tag{40}$$

An invalid newest report postpones learning; an older version is never substituted. Use the report at actual consumption time ι_t, even if a missing forecast delayed it beyond maturity. Each row is learned at most once; subsequent revisions are ignored. The report used for a consumed row is

$$\bar{y}_s^{(d)} = y_s^{[\iota_{\tau_d(s)}]}, \qquad \tau_d(s) \text{ exists}. \tag{41}$$

Plain MultiQT on counts starts at $\tilde{\theta}_0 = 0$ and, once active, updates against previously returned forecasts before computing q_t:

$$\tilde{\theta}_t^{\alpha} = \tilde{\theta}_{t-1}^{\alpha} - \eta_t \sum_{s \in \mathcal{R}_t^{(d)}} (\mathbf{1}\{y_s^{[\iota_t]} \leq q_s^{\alpha}\} - \alpha), \tag{42}$$

$$\theta_t = \Pi_{\mathcal{K}-b_t}(\tilde{\theta}_t), \qquad q_t = b_t + \theta_t. \tag{43}$$

The hidden state is unprojected; an empty release set gives no update. The `log1p` tracker applies the same rule in ψ coordinates: ψ is increasing, so the coverage events and the update (42) are unchanged, with $\tilde{\theta}_t$ in log units, and the forecast is

$$q_t = \psi^{-1}\left[\Pi_{\mathcal{K}}(\beta_t + \tilde{\theta}_t)\right], \qquad \psi(x) = \log(1+x), \quad \psi^{-1}(z) = e^z - 1. \tag{44}$$

The projection acts in log units, before the inversion. Forecasts stay above -1; as with the `identity` tracker, they are not clipped at zero.

Weekly re-run. For forecasts and states, a superscript $[t]$ identifies the re-run at reference week t: $q_s^{[t]}$ is its reconstructed forecast for row s. This differs from the report cutoff date v in $y_s^{[v]}$. At reference week t, reconstruct plain MultiQT from zero using reports current at ι_t, retaining the release positions $j = \tau_d(s)$ and delay d, but omitting invalid current reports:

$$\tilde{\theta}_j^{\alpha,[t]} = \tilde{\theta}_{j-1}^{\alpha,[t]} - \eta_j^{[t]} \sum_{\substack{s \in \mathcal{R}_j^{(d)} \\ y_s^{[\iota_t]} \text{ valid}}} (\mathbf{1}\{y_s^{[\iota_t]} \leq q_s^{\alpha,[t]}\} - \alpha), \tag{45}$$

$$\theta_j^{[t]} = \Pi_{\mathcal{K}-b_j}(\tilde{\theta}_j^{[t]}), \qquad q_j^{[t]} = b_j + \theta_j^{[t]}. \tag{46}$$

Only $q_t^{[t]}$ is returned; earlier returned forecasts are unchanged. Reference scales follow Section 3.3. Studentized methods use (22), reconstruct their scale history and return the base until a required scale exists.

Report filtering. Select the latest report at the cutoff, then apply the reviewed exclusions; never substitute an older or future report. Count the delay from the first acceptable report. Stateful updates wait for an acceptable current report; re-runs omit excluded versions. Use the latest version for scoring and peak detection if acceptable; otherwise leave that target unscored for every method. Apply the same exclusions to the training reports of our own base forecasts.

3.3 Tuning

Reference scales use the first 20 consumed rows in release order, $I_{20}^{(d)}$, which need not span 20 calendar weeks. Parameter selection uses the tuning period; these are separate operations.

Reference scales and retrospective initialization. Write Q_p for an empirical p-quantile. The initial error scale S_0, distinct from the score function S_t, pools absolute base errors in tracker units:

$$S_0 = \max\Big\{\varepsilon_S, Q_{0.9}\Big(\Big\{|\psi(b_s^\alpha) - \psi(\bar{y}_s^{(d)})| : s \in I_{20}^{(d)},\ \alpha \in \mathcal{A}\Big\}\Big)\Big\}, \qquad \varepsilon_S = 10^{-6}. \tag{47}$$

It sets rates $\eta = \kappa S_0$, with rate multiplier κ, and studentized scale bounds $[.05S_0, S_0]$; studentized step sizes are selected in standardized units. Script 02 labels the multiplier m, distinct from $m = |\mathcal{A}|$ here. The reference loss L_0 is defined in Section 3.5.3.4.

With the default `replay_warmup=TRUE`, both constants use the eventual first 20 releases and apply from the start. Forecasts before the 20th release therefore use retrospective scale information. With replay disabled, learning waits for warmup; fewer than 20 releases leaves learning inactive. Estimate both constants from the stateful feedback schedule and keep them fixed across weekly re-runs; states, weights and time-varying scales are reconstructed. Evaluation never resets the tracker, so its initial evaluated correction is generally nonzero.

Cases and selection. Tune COVID and flu individually, each with the `log1p` tracker (`fixed` and `leaky` only) before the `identity` tracker (every tuned method). For each disease and tracker, run stateful delay 0, delay 1 and weekly re-run at delay 0. Each of these 12 configurations has one tracker-rate tuning stage, starting from defaults. Configurations without tuning rows retain the defaults. Tune all horizons of the base: $-1, \ldots, 3$ with hub forecasts, $0, \ldots, 3$ with our own base forecasts.

Candidates replay the full history with common S_0, L_0 within each case, tracker and feedback design. Only tuning target weeks are scored, including initialization rows when valid. Scores use latest archived truth, which need not have been available before evaluation. Pool case-level means equally over non-US locations and horizons. The criterion is log quantile loss for `log1p` and quantile loss per 100,000 for `identity`, without L_0 normalization.

Select tracker rates separately for `fixed`, `leaky`, `studentized`, `studentized_base`, `median_constant`, `studentized_full` and `studentized_level_rms`. `fixed_log` and `leaky_log` use the rates selected for `fixed` and `leaky` with the `log1p` tracker, that is by log quantile loss.

Let $\overline{\mathcal{L}}_j(\vartheta)$ be the mean loss of method j at candidate ϑ over tuning cases, and let $\overline{\mathrm{CE}}_j(\vartheta)$ be its mean calibration error. Subscript 0 denotes the original forecast. For each method, minimize the sum of percentage changes from the original:

$$E_j(\vartheta) = 100\left[\frac{\overline{\mathcal{L}}_j(\vartheta)}{\overline{\mathcal{L}}_0} - 1\right] + 100\left[\frac{\overline{\mathrm{CE}}_j(\vartheta)}{\overline{\mathrm{CE}}_0} - 1\right]. \tag{48}$$

The code calls this the elbow rule: one percentage point of loss change and one percentage point of calibration-error change have equal weight. It does not estimate curvature. Ties select the smaller parameter value. Each method has one tuned rate, so its grid is evaluated separately; there is no Cartesian search.

Setting	Candidates
`fixed_multiplier`	.001, .0025, .005, .01, .02, .05, .1, .2, .4
`median_rate`	.003, .01, .03, .1, .3, 1
`leaky_multiplier`	.001, .002, .004, .008, .02, .05, .1, .2, .3, .5, 1

Multipliers scale S_0. `median_rate` is standardized.

Fixed and inherited settings. All learned mixtures use fixed $H = 8$ processed weight updates; sharing is therefore $\gamma = 1/(2H) = 1/16$ where enabled. Learned mixtures use adaptive λ from $\lambda_{\text{start}} = 2.72$; $\lambda = 0$ gives equal weights in `exp_weights()`. Recent-start count is 8; the two-rate recent starts use $.03S_0$ and $.3S_0$; the outer rate is $.001S_0$. Other constants remain as specified with each method. `studentized_zero` and `median_constant_zero` inherit their corresponding normal-start rates, and outer variants retain $H = 8$ of their inner mixtures.

Tuning plots. The current `multiqt_tuning.pdf` pairs calibration error with quantile loss, both as percentage changes from the original. Solid/dashed lines use the log/identity criteria. Symbol size grows with the candidate value; circles mark selected values and an open square at $(0, 0)$ marks the original. A dotted line of slope -1 through each selected point shows its value of (48); no candidate from that curve lies below it. Only pooled pages by disease and feedback design are produced, using the non-US locations and tuning horizons above. The `on_grid_edge` flag marks an endpoint selection, not evidence of an interior optimum.

3.4 Evaluation

Peaks and evaluation windows. Let $y_t^\star$ be the latest archived count for the target of row t, and $y^\star(D)$ that of target week D, so $y_t^\star = y^\star(T_t)$ for the target date T_t of row t. Peaks are found per disease and location. Script 01 omits invalid latest counts, including reviewed exclusions, and the incomplete-reporting weeks 4 May–2 November 2024. Write $D_1 < \cdots < D_n$ for the retained target weeks, $x_i = y^\star(D_i)$ for their counts and $\mathcal{N}_i(k) = \{j \in \{1, \ldots, n\} : |j - i| \le k\}$ for the retained observations within k of i. Indices count retained observations, not calendar weeks: across the omitted weeks, neighbours D_i, D_{i+1} can be months apart. `find_peaks()` then uses three steps:

1. *Smoothing and candidates.* The centred three-observation mean, with the endpoints unchanged, is
$$\mu_i = \tfrac{1}{3}(x_{i-1} + x_i + x_{i+1}), \quad 1 < i < n, \qquad \mu_1 = x_1, \quad \mu_n = x_n. \tag{49}$$
Observation i is a candidate if μ_i is the largest smoothed value within eight observations on either side, the earliest at ties:
$$\mu_i \ge \mu_j \text{ for all } j \in \mathcal{N}_i(8), \qquad \mu_i > \mu_j \text{ for all } j \in \mathcal{N}_i(8) \text{ with } j < i. \tag{50}$$

2. *Prominence.* On each side, search up to the nearest strictly higher smoothed value, or the series end:
$$l_i = \max(\{1\} \cup \{j < i : \mu_j > \mu_i\}), \qquad r_i = \min(\{n\} \cup \{j > i : \mu_j > \mu_i\}). \tag{51}$$
The prominence is the smoothed height above the higher of the two minima in these ranges,
$$\pi_i = \mu_i - \max\Big\{\min_{l_i \le j \le i} \mu_j,\ \min_{i \le j \le r_i} \mu_j\Big\} \ge 0, \tag{52}$$
so $\pi_1 = \pi_n = 0$. A candidate is kept if
$$\pi_i > 0, \qquad \pi_i \ge 0.3\,\mu_i, \qquad \pi_i \ge 0.1\,M, \tag{53}$$
where $M = \max\{x_j : D_j \text{ is a forecast target week}\}$, over the forecast target weeks of all horizons. Equivalently, on both sides the smoothed counts fall to $\min\{0.7\,\mu_i,\ \mu_i - 0.1M\}$ or below before they exceed μ_i or the series ends. Script 01 stores π_i/μ_i as `relative_prominence`.

3. *Placement.* A kept candidate i moves to the largest raw count within one observation, the earliest at ties:
$$i' = \min\Big\{\arg\max_{j \in \mathcal{N}_i(1)} x_j\Big\}. \tag{54}$$
Its peak date $D_{i'}$ is kept if it lies between the first and the last forecast target week.

Peaks are hindsight labels, used for prediction only by the target oracle.

Let $p(t)$ be the peak date $D_{i'}$ nearest to T_t (earlier at ties), and $(T_t - p(t))/7$ their distance in calendar weeks. For evaluation rows E with valid latest truth, score the windows

$$\begin{aligned} W_{\text{overall}} &= E, \\ W_{\text{before}} &= \{t \in E : -7 \le (T_t - p(t))/7 \le 0\}, \\ W_{\text{after}} &= \{t \in E : 1 \le (T_t - p(t))/7 \le 8\}. \end{aligned} \tag{55}$$

Rows count once; the peak week belongs to "up to peak". Overall also includes weeks outside these windows. Missing rows and period boundaries shorten windows without discarding peaks. Period membership follows the target date; its nearest peak may lie in another period. The main plots use these windows over the full evaluation period.

Loss and calibration error. A case fixes location, horizon and feedback design within a disease. For a selected window W of $N = |W|$ rows, average over rows and all 23 quantile levels:

$$\mathcal{L}^{\text{nat}}(W) = \frac{1}{mN} \sum_{t \in W} \sum_{\alpha \in \mathcal{A}} \rho_\alpha(q_t^\alpha / c_{\text{scale}}, y_t^\star / c_{\text{scale}}), \tag{56}$$

$$\mathcal{L}^{\log}(W) = \frac{1}{mN} \sum_{t \in W} \sum_{\alpha \in \mathcal{A}} \rho_\alpha(\log(1 + \max\{q_t^\alpha, 0\}), \log(1 + y_t^\star)), \tag{57}$$

$$\text{CE}(W) = \frac{1}{m} \sum_{\alpha \in \mathcal{A}} \left| \frac{1}{N} \sum_{t \in W} \mathbf{1}\{y_t^\star \le q_t^\alpha\} - \alpha \right|. \tag{58}$$

Columns are `quantile_loss`, `log_quantile_loss` and `coverage_error`; tuning uses `loss`, `log_loss` and `coverage_error`. CE averages weeks before taking the absolute residual. All feedback designs use the same latest truth; empty windows give missing scores. Evaluation losses are never divided by L_0. Scripts 04/06 display the criterion of the tracker; script 02 displays loss per 100,000.

All-location trade-offs. Write $\mathcal{L}_{j,c}$ and $\text{CE}_{j,c}$ for the mean quantile loss and calibration error of method j in case c; $j = 0$ is the original. A set $\mathcal{C}$ contains J cases. In `multiqt_peak_reasoning_tradeoff.pdf`, columns are horizons $-1, \ldots, 3$; blocks are delay 0, delay 1 and re-run, each showing overall, up to peak and after peak. Lower left is better. The first page per disease shows individual non-US cases and method means

$$\left(J^{-1} \sum_c \mathcal{L}_{j,c},\ J^{-1} \sum_c \text{CE}_{j,c} \right).$$

These average location scores, not forecasts or pooled coverage indicators. The first row of `multiqt_error_tradeoff.pdf` uses the same aggregation for fixed rates. Feedback designs have separately selected parameters, so their differences include retuning. Axes vary by panel; all-location limits use the 1st–99th percentiles and mean points, with loss capped near four times the base. Extremes can be hidden or marked `>`. Original/common-weight/outer methods use open circles/triangles/squares or diamonds; exact forecast duplicates are hidden.

Evaluation tree. Script 06 splits pooled disease cases by feedback design, then horizon. For each window it reports

$$R_j(\mathcal{C}) = 100 \left[1 - \frac{\sum_{c \in \mathcal{C}} N_c \mathcal{L}_{j,c}}{\sum_{c \in \mathcal{C}} N_c \mathcal{L}_{0,c}} \right], \tag{59}$$

$$P_j(\mathcal{C}) = J^{-1} \sum_{c \in \mathcal{C}} \mathbf{1}\{\mathcal{L}_{j,c} < \mathcal{L}_{0,c}\}. \tag{60}$$

`reduction` is a ratio of week-weighted losses (positive is better); `share_improved` weights cases equally, counts ties as not improved, and is printed for at least five cases. CE panels also average case scores with week weights. Each page displays the ten methods with lowest pooled after-peak loss: a descriptive evaluation ranking, not tuning. Tree panels share axes; clipping is marked < or >. By contrast, script 02 phase-rate plots show loss change $100(\mathcal{L}_j/\mathcal{L}_0 - 1)$, where negative is better.

Forecast curves. Files in `multiqt_method_forecasts/` show levels $.1, \ldots, .9$ and weekly loss per 100,000 at a fixed horizon; gray marks tuning, dotted lines peaks. Files in `multiqt_season_forecasts/` add coverage and loss trajectories for selected fixed rates. `multiqt_base_forecast_paths.pdf` shows full count timelines, median paths around all retained peaks and three diagnostic indicators (Section **??**). It spans tuning and evaluation; its flu winter 2024/25 peak is within tuning, not held-out evidence. Weight timelines show how each mixture combines its experts. Gap dates are listed in Section 3.1.

3.5 Methods

The methods below use the preceding data, feedback, tuning and evaluation conventions unless stated otherwise.

Methods not run. Since 13 November 2026, 28 of the 52 methods stay in the code but are not run (`paused_methods` in `R/03_functions.R`; deleting a name runs the method again), chosen from the run of 12 November (Section 4): the median rebuilds `studentized`, `studentized_zero`, `studentized_base`, `median_constant` and `median_constant_zero` (+37% to +60%; their question is answered in Section 4.6) and `studentized_full`; the t-PIT scale with its four mixtures, until its explosion is fixed (Section 4.8); the equal-weight rate grids and equal-weight recent starts of the power and spread scales (no better than the learned ones); the rate grids with sharing and the outer MultiQT (`shared`, `shared_common`, `shared_outer`, `shared_common_outer`; sharing costs 5 to 6 points); `restart_persistent`, `restart_weighted` and the outer weekly starts (`restart_uniform` and `restart_recent` keep the comparison of all starts with the eight youngest); `score_equal` (alike `score_learned`); `oracle_target` (its point is made); and `leaky_log`, `fixed_log_adaptive` and `sqrt_adaptive`, which lie between other trackers. The 23 methods run besides the base are Fixed and Leaky MultiQT with their adaptive versions, REF-op, `sqrt_constant`, `fixed_log`, `leaky_log_adaptive`, `studentized_level_rms`, `studentized_pit`, the power and spread trackers with their learned rate grids and recent starts, `uniform`, `weighted`, `weighted_common`, `restart_uniform`, `restart_recent`, `restart_recent_dual` and `score_learned`. The results of Section 4 still include the methods not run; the next run will not.

3.5.1 Single trackers

These methods maintain one MultiQT tracker for each case.

Fixed rate: `fixed`. Use (42)–(43) with $\eta = \kappa S_0$ in tracker units. Tune κ separately (default .05). Corrections persist across rounds.

Leaky tracker: `leaky`. Use Delphi's leak (`fast_decay` of their tracker implementation [23]): Fixed MultiQT whose hidden state loses the fraction $\delta = .1$ at every processed reference week, before its update,

$$\tilde{\theta}_t^{\alpha} = (1-\delta)\tilde{\theta}_{t-1}^{\alpha} - \eta_t \sum_{s \in \mathcal{R}_t^{(d)}} \left(\mathbf{1}\{y_s^{[\iota_t]} \le q_s^{\alpha}\} - \alpha\right), \qquad \eta_t = \kappa S_0, \tag{61}$$

with the forecast of (43). A correction that new feedback does not renew fades back to the base, with a half-life of about 6.6 reference weeks; the decay applies even without feedback, and gaps do not compound it. This is the fast term of `ref_op` without its slow term and rolling rate. Tune κ separately; δ is Delphi's value and is not tuned. The method also belongs to the score bank.

Log1p trackers: `fixed_log` and `leaky_log`. Fixed and Leaky MultiQT on the `log1p` tracker, $\psi(x) = \log(1+x)$: the updates (42) and (61) of the hidden state $\tilde{\theta}_t$, in log units, and the forecast (44). The rate is $\eta = \kappa S_0$ with S_0 from (47) in log units (in a re-run, frozen from the stateful design, as for the other methods), and κ is the rate selected for `fixed` and `leaky` with the `log1p` tracker. Their correction is multiplicative, (70): they compare an additive correction with one that scales with the level (Section 3.5.2).

Persistent restart: `restart_persistent`. Keep the first eligible weekly-start expert with $\kappa = .05$ (Section 3.5.3.5). It learns only from forecasts made since its start. Its rate is not separately tuned, so it need not reproduce `fixed`.

3.5.2 Studentized scores

Studentized variants and median initialization. Studentized methods track counts. Use $a_t^\alpha = b_t^\alpha$ for the full base or $a_t^\alpha = b_t^5$ for its median. After processing available feedback, return

$$q_t = \Pi_{\mathcal{K}}^{A_t^{-2}}(a_t + A_t \tilde{u}_t). \tag{62}$$

For a common scale, the projection is the ordinary isotonic projection. After warmup, the delayed update is applied to the hidden state,

$$\tilde{u}_t^\alpha = \tilde{u}_{t-1}^\alpha + \eta \sum_{s \in \mathcal{R}_t^{(d)}} \left(\alpha - \mathbf{1}\{\bar{y}_s^{(d)} \le q_s^\alpha\}\right). \tag{63}$$

Compare each report with its historical forecast q_s, reconstructed inside a re-run when applicable. Project only the forecast; the hidden state carries forward throughout the run. The standardized step has default $\eta = .1$ and is tuned for each normal-start or full-base method. Zero-start variants inherit the corresponding normal-start rate.

Scales from recent errors or the current forecast. Let $\mathcal{H}_t^{(d)}$ contain the last five consumed rows, or fewer if unavailable, and set $e_s^\alpha = \bar{y}_s^{(d)} - b_s^\alpha$. These need not be consecutive calendar weeks. A hat marks a raw scale estimate, and Φ is the standard normal distribution function. The default scales are

$$\widehat{\sigma}_{t,\text{med}} = \left[\frac{1}{|\mathcal{H}_t^{(d)}|} \sum_{s \in \mathcal{H}_t^{(d)}} (e_s^{.5})^2\right]^{1/2}, \tag{64}$$

$$\widehat{\sigma}_{t,\text{res}}^\alpha = \left[\frac{1}{|\mathcal{H}_t^{(d)}|} \sum_{s \in \mathcal{H}_t^{(d)}} (e_s^\alpha)^2\right]^{1/2}. \tag{65}$$

Both require five consumed rows. RMS includes bias because the errors are not centred. The optional robust alternative is median $|e_s^\alpha|/\Phi^{-1}(.75)$.

The common scale inferred from the current base is

$$\widehat{\sigma}_{t,\text{base}} = \frac{b_t^{.9} - b_t^{.1}}{\Phi^{-1}(.9) - \Phi^{-1}(.1)}. \tag{66}$$

The denominator is approximately 2.563. Base-width scales respond immediately to changes in the current forecast; residual scales respond only after the corresponding reports arrive.

Floor raw scales at 10^{-6}, then clip to $[.05S_0, S_0]$ once S_0 exists; denote these bounded scales by σ. The bounds limit how much the scale suppresses or amplifies accumulated corrections. They are implementation defaults, not requirements of studentization.

Scales of the most stationary scores. Three further studentized methods use the scales whose scores drifted least (Section 10), chosen from three different ideas: the forecast's level, where the forecast lies among the recent reports, and the forecast's own spread. Each is a function of the forecast and of reports known at the cutoff:

$$\widehat{\sigma}_{t,\mathrm{pow}} = (1 + b_t^{.5})^p, \qquad p = .5, \tag{67}$$

$$\widehat{\sigma}_{t,\mathrm{tpit}} = \frac{g_\nu(G_\nu^{-1}(\hat{F}_t(b_t^{.5})))}{\hat{F}_t'(b_t^{.5})}, \qquad \nu = 25, \tag{68}$$

$$\widehat{\sigma}^{\alpha}_{t,\mathrm{spr}} = s_t\Big(1 + \frac{|b_t^{\alpha} - b_t^{.5}|}{s_t}\Big)^{1-\lambda}, \qquad s_t = \tfrac{1}{2}(b_t^{.75} - b_t^{.25}),\ \lambda = .4. \tag{69}$$

The power scale (`power_sigma()`) grows with the forecast median. The t-PIT scale (`tpit_sigma()`) is the derivative scale of the t-PIT of [21] at the forecast median: $\hat{F}_t$ is the distribution function of the last 26 reports known at the cutoff (at least 8), smoothed by a normal kernel, and G_ν the t distribution with density g_ν. The spread scale (`spread_sigma()`) is the derivative scale of a symmetric Box-Cox transform centred at the forecast median: per level, U-shaped around the median, with the weighted projection. Each scale is kept within a factor of 20 of the median of its values so far, the current week included (the t-PIT scale explodes when the forecast median leaves the range of the recent reports), then divided by its median over the S_0 rows and multiplied by S_0 (`normalized_scale()`), so a rate m in units of the scale acts like mS_0 in a typical week and the rate grid and the restart rate of the unscaled banks carry over; the bounds $[.05S_0, S_0]$ would cut the peaks of these scales and are not applied. Each scale serves a tracker (`studentized_power`, `studentized_tpit`, `studentized_spread`, rate tuned as for the other studentized methods) and four mixtures (Table 4). Added on 12 November 2026, tuned and run the same evening (Section 4.8). The guard of 20 does not stop the t-PIT scale from exploding (Section 4.8, "Known issue").

Table 2: Studentized methods. Scales refer to the bounded versions above. Initial states are specified before the first update.

Method ID	Centre and scale	Initialization and purpose
`studentized`	$a_t^{\alpha} = b_t^{.5}$; common $\sigma_{t,\mathrm{med}}$	$\tilde{u}_{\mathrm{init}}^{\alpha} = \Phi^{-1}(\alpha)$. Reconstruct the distribution around the base median using recent errors.
`studentized_base`	$a_t^{\alpha} = b_t^{.5}$; common $\sigma_{t,\mathrm{base}}$	$\tilde{u}_{\mathrm{init}}^{\alpha} = \Phi^{-1}(\alpha)$. Reconstruct the distribution from the median and the central width of the current base forecast.
`median_constant`	$a_t^{\alpha} = b_t^{.5}$; common constant σ	$\tilde{u}_{\mathrm{init}}^{\alpha} = \Phi^{-1}(\alpha)$; freeze $\sigma_{t,\mathrm{med}}$ at the first reference week after both required warmups. Separates replacing the base by its median from subsequently changing the scale.
`studentized_zero`	$a_t^{\alpha} = b_t^{.5}$; common $\sigma_{t,\mathrm{med}}$	$\tilde{u}_{\mathrm{init}} = 0$. Start all levels at the median; use the rate selected for `studentized`.

Continued on next page

Table 2 (continued)

Method ID	Centre and scale	Initialization and purpose
`median_constant_zero`	$a_t^\alpha = b_t^{5}$; common constant σ	$\tilde{u}_{\text{init}} = 0$; use the scale and rate of `median_constant`.
`studentized_full`	$a_t^\alpha = b_t^\alpha$; common $\sigma_{t,\text{base}}$	$\tilde{u}_{\text{init}} = 0$. Keep the full base and rescale its accumulated corrections together.
`studentized_level_rms`	$a_t^\alpha = b_t^\alpha$; $\sigma_{t,\text{res}}^\alpha$	$\tilde{u}_{\text{init}} = 0$. Set the scale at each level from recent errors at that level; weighted isotonic projection.
`studentized_power`	$a_t^\alpha = b_t^\alpha$; common $\sigma_{t,\text{pow}}$ (67)	$\tilde{u}_{\text{init}} = 0$. Corrections scale with the forecast median.
`studentized_tpit`	$a_t^\alpha = b_t^\alpha$; common $\sigma_{t,\text{tpit}}$ (68)	$\tilde{u}_{\text{init}} = 0$. Corrections scale with where the forecast median lies among the recent reports.
`studentized_spread`	$a_t^\alpha = b_t^\alpha$; $\sigma_{t,\text{spr}}^\alpha$ (69)	$\tilde{u}_{\text{init}} = 0$. Per-level spread of the forecast; weighted isotonic projection.

The scale of `median_constant` is fixed within each reconstructed history but can change between re-runs, unlike the reference scales (Section 3.3). Zero-start variants compare initialization at the same selected rate. Their first active forecast may already include a feedback update.

The residual-median methods also allow $t_5^{-1}(\alpha)$ initialization; the base-width median method always uses normal quantiles. It retains the median and bounded central width but imposes a symmetric normal shape in counts, so it need not reproduce the original forecast. Until a required residual scale exists, return the original full base. Feedback consumed while waiting is not replayed; later feedback on those rounds is compared with the original forecasts.

Studentization rescales the accumulated correction, rather than only the newest increment. Quantile-specific scales retain the coverage target, but no calibration guarantee is established here for those variants.

Relation to the `log1p` tracker. Standard MultiQT on the `log1p` tracker (`fixed_log`, `leaky_log`) nevertheless acts like a studentized method whose scale is the level. Its base in tracker units is $\beta_t^\alpha = \psi(b_t^\alpha) = \log(1 + b_t^\alpha)$. Without crossings, (44) gives $q_t^\alpha = \psi^{-1}(\beta_t^\alpha + \tilde{\theta}_t^\alpha)$, that is,

$$1 + q_t^\alpha = (1 + b_t^\alpha)\, e^{\tilde{\theta}_t^\alpha}, \qquad q_t^\alpha - b_t^\alpha = (1 + b_t^\alpha)(e^{\tilde{\theta}_t^\alpha} - 1) \approx (1 + b_t^\alpha)\, \tilde{\theta}_t^\alpha. \tag{70}$$

The state does not add counts to the base: it scales the shifted base $1 + b_t^\alpha$ by the factor $e^{\tilde{\theta}_t^\alpha}$ (the shift by one only keeps zero counts finite on the log scale). The correction in counts is therefore proportional to the current level.. The `identity` tracker, (43), adds the same number of counts at every level.

The state is the threshold for the residual on the log scale: level α is covered if $\psi(y) - \beta_t^\alpha \le \tilde{\theta}_t^\alpha$ (without crossings). This residual is measured relative to the level. Exactly,

$$\psi(y) - \beta_t^\alpha = \log\Big(1 + \frac{y - b_t^\alpha}{1 + b_t^\alpha}\Big) \approx \frac{y - b_t^\alpha}{1 + b_t^\alpha}, \tag{71}$$

with $\log(1 + x) \approx x$ for small relative errors x. A log-scale residual is thus approximately a count residual divided by the level: the studentized score (13) with $\sigma_t^\alpha = 1 + b_t^\alpha$, with $\tilde{\theta}_t^\alpha$ in the role of $\tilde{u}_t^\alpha$ in (14). The approximation only interprets the scale; the tracker uses the exact coverage events and (44). Like $\sigma_{t,\text{base}}$, this scale is known at the cutoff, with no feedback lag, and a change of it rescales the whole accumulated correction (Section 2.1.5).

In general, a tracker transform studentizes by the reciprocal of its derivative at the base, since $\psi(y) - \beta_t^\alpha \approx \psi'(b_t^\alpha)(y - b_t^\alpha)$, and likewise for χ: $1/\psi'(b_t^\alpha) = 1 + b_t^\alpha$ for `log1p` and one

for `identity` (the additive correction (43) of `fixed` and `leaky`); for the square-root trackers χ of `ref_op` and `sqrt_constant` (Section 3.5.4), $1/\chi'(b_t^\alpha)$ is a multiple of $\sqrt{b_t^\alpha}$, in between. Section 4.5.1 shows what level-tied scales do after peaks.

Scores and scales tested. Table 3 lists every score of the base that was tested for a stable α-quantile (Section 2.1.7), whether a method uses it, and where it came from. A score is a transform $\psi_t(y) - \psi_t(b_t^\alpha)$ (28), which always keeps inward flow, or a residual divided by a scale (13), common to all levels (always inward flow) or one per level (only inside the band (27)). `R/07_score_selection.R` computes all of them (`score_candidates` and `candidate_scores()` in `R/03_functions.R`) and compares them by the drift of their hindsight threshold, by their peak-aligned coverage and by the running quantile (Sections 4.5.2 and 4.5.3). The candidates come from six sources.

Trackers in use. Standard MultiQT corrects additively, the identity score. Delphi's production trackers work on square-root counts (Section 3.5.4), and Delphi's calibration experiments also used `log1p`: for counts whose variance grows with their level, these scales make errors at high and low incidence comparable (Section 3.5.2).

Recent errors. The first studentized methods divide the residual by the spread of the last five errors, per level (level RMS, (65)) or around the median with normal quantiles (median rebuild, (64)). They learn the size of the errors from the reports and so lag the turn of a wave by the feedback delay. The level RMS scales projected into the band (`studentized_level_band`) drifted like the unprojected ones; that method was removed on 12 November 2026.

Spread of the forecast. The base's own width is known at the cutoff, without feedback lag: its central 80% interval in normal units (base width, (66)), half its interquartile range, and the spread per level (69). The PIT of the base expresses the outcome in the base's own probability units.

Level of the forecast. Errors grow with incidence. The Box-Cox family on counts, ψ_λ (Section 2.1.6), runs from the identity ($\lambda = 1$) through the square root (up to a shift, $\lambda = .5$) to `log1p` ($\lambda = 0$). Its derivative at the forecast median is the scale $(1 + b_t^{.5})^p$ with $p = 1 - \lambda$: a power of the forecast median, tested for p from -1 to 2, per level and as the exact transform.

Revisions of the forecast. The forecasts of one target week at successive cutoffs, that is at different horizons, disagree roughly as much as the base errs. The revision scale is the median absolute difference between this cutoff's median for a target week and the medians issued for it at earlier cutoffs, over every target of the cutoff or only the row's own target (none at the longest horizon): a scale from the forecasts alone, known at the cutoff (`revision_sigma()`). The method `studentized_revision` was removed on 12 November 2026.

Transforms of Uniejewski (2025). Uniejewski [21] standardizes the series by the median and the median absolute deviation of the calibration window, $z = (x - m)/s$, and applies one of: the parametrized asinh $\operatorname{sgn}(z)[\operatorname{asinh}(|z| + a) - \operatorname{asinh}(a)]$, $a = \sqrt{1/c^2 - 1}$, with slope c at the origin ($c = 1$: asinh); a symmetric Box-Cox $\operatorname{sgn}(z)\{(|z| + 1)^\lambda - 1\}/\lambda$; the mirror log $\operatorname{sgn}(z)\log(1 + c|z|)$; or a PIT with a t distribution, $G_\nu^{-1}(\hat{F}(x))$. Here m and s come from the last 26 reports known at the cutoff ("window", as in the paper; z_w below) or from the forecast, the median $b_t^{.5}$ and half the base interquartile range s_t ("centred"; z_c). With m and s fixed at the cutoff, each $\psi_t(y) = T(z(y))$ is strictly increasing, so all are scores of the form (28). Their derivatives give scales: at the forecast median one common scale $s/T'(z(b_t^{.5}))$; per level U-shaped

around m, $\sqrt{s^2 + (b_t^\alpha - m)^2}$ for asinh, $s(1 + |z|)^{1-\lambda}$ for Box-Cox and $s/c + |b_t^\alpha - m|$ for the mirror log. Centred at the forecast median, every derivative scale at the median is proportional to s_t. The empirical distribution function of the window is only nondecreasing, flat between and beyond the reports; smoothed with a normal kernel, as in `tpit_sigma()`, it has a derivative, and the t-PIT scale at the median measures where the forecast median lies among the recent reports. The paper's parameter grids (asinh $c = .1, \ldots, 1$, Box-Cox $\lambda = 0, \ldots, 1$, mirror log $c = .1, \ldots, 1$, t-PIT $\nu = 9, \ldots, 49$, each exact, at the median and per level) were explored first (`agent/scripts/residual_scales.R`); the parameter changed the drift by at most 0.03, and R/07 keeps one or two members of each.

Table 3: Scores and scales tested. Kind: transform (inward flow always), common scale (always) or scale per level (only in the band (27)). Tested: D drift ratio and peak-aligned coverage, R running quantile (both R/07), M MultiQT with the finalist scales (Section 4.5.4), W a method of R/04. $z_w(x) = (x - m_w)/s_w$ with the median m_w and median absolute deviation s_w of the last 26 reports; $z_c(x) = (x - b_t^{.5})/s_t$ with half the base interquartile range s_t.

Score or scale	$S_t^\alpha(y)$	Kind	Tested	Methods
Trackers in use				
Identity	$y - b_t^\alpha$	transform	D R M W	`fixed`, `leaky`
Square root	$\sqrt{y} - \sqrt{b_t^\alpha}$	transform	D R W	`ref_op`, `sqrt_constant`
`log1p`	$\log(1 + y) - \log(1 + b_t^\alpha)$	transform	D R W	`fixed_log`, `leaky_log`
Level of the forecast				
Box-Cox on counts, $\lambda = .25, .75$	$\psi_\lambda(y) - \psi_\lambda(b_t^\alpha)$	transform	D R	—
Power of the median, $p = -1, .25, .5, .75, 1, 1.5, 2$	$(y - b_t^\alpha)/(1 + b_t^{.5})^p$	common scale	D R M W	`studentized_power` ($p = .5$) and its mixtures
Power per level, $p = .75$	$(y - b_t^\alpha)/(1 + b_t^\alpha)^p$	per level	D R	—
Recent errors				
Level RMS	$(y - b_t^\alpha)/\sigma_{t,\text{res}}^\alpha$	per level	D R W	`studentized_level_rms`
Level RMS in the band	scales $\sigma_{t,\text{res}}^\alpha$ projected onto (27)	per level	D (earlier)	`studentized_level_band`, removed
Median rebuild	$(y - b_t^{.5})/\sigma_{t,\text{med}} - \Phi^{-1}(\alpha)$	common scale	D R W	`studentized`, `studentized_zero` (not run)
Spread of the forecast				
Base width	$(y - b_t^\alpha)/\sigma_{t,\text{base}}$	common scale	D R M W	`studentized_full`, `studentized_base` (not run)
Half the base IQR	$(y - b_t^\alpha)/s_t$	common scale	D R	—
Spread per level	$(y - b_t^\alpha)/\sigma_{t,\text{spr}}^\alpha$	per level, U-shaped	D R W	`studentized_spread` and its mixtures
PIT of the base	$F_{b_t}(y) - \alpha$	transform	D R W	`studentized_pit`
Revisions of the forecast				
Revisions, every target or own target	$(y - b_t^\alpha)/\sigma_{t,\text{rev}}$	common scale	D R	`studentized_revision`, removed
Transforms of Uniejewski (2025)				
asinh ($c = 1$), window, exact	$T(z_w(y)) - T(z_w(b_t^\alpha))$	transform	D R M	—

Continued on next page

Table 3 (continued)

Score or scale	$S_t^\alpha(y)$	Kind	Tested	Methods
asinh ($c = 1$), Box-Cox ($\lambda = 0$), mirror log ($c = .3$): window, at the median	$(y - b_t^\alpha)\, T'(z_w(b_t^{.5}))/s_w$	common scale	D R M	—
t-PIT ($\nu = 25, 49$), window, at the median	$(y - b_t^\alpha)/\sigma_{t,\text{tpit}}$	common scale	D R W	`studentized_tpit` ($\nu = 25$; not run, explodes)
asinh ($c = 1$), mirror log ($c = 1/3$): centred, exact	$T(z_c(y)) - T(z_c(b_t^\alpha))$	transform	D R	—

3.5.3 Weighted experts

An expert is a complete forecasting procedure; a bank is the set being combined. Weights learn from quantile losses separately for each location–horizon case. An adaptive rate changes their responsiveness.

3.5.3.1 Mixtures and feedback losses For K experts, let $f_{k,t}^\alpha$ be the forecast of expert k and w_k^α its weight. Suppressing time indices for one forecast, form the mixture $\hat{q}$ and repair any crossings:

$$\hat{q}^\alpha = \sum_{k=1}^{K} w_k^\alpha f_k^\alpha, \qquad w_k^\alpha \geq 0, \quad \sum_{k=1}^{K} w_k^\alpha = 1, \qquad q = \Pi_{\mathcal{K}}(\hat{q}). \tag{72}$$

Per-quantile weights can introduce crossings, repaired by isotonic projection. Common weights replace w_k^α by w_k and preserve order when every expert is ordered. All mixtures average counts.

Let n count processed feedback updates, with forecast row $s(n)$ and its report $\bar{y}_{s(n)}^{(d)}$ (41); in a re-run at reference week t, the report current then, $y_{s(n)}^{[\iota_t]}$, takes its place. Score the expert forecasts for that row, using L for unnormalized losses and ℓ for losses normalized by L_0:

$$L_{k,n}^\alpha = \rho_\alpha\left(f_{k,s(n)}^\alpha / c_{\text{scale}}, \bar{y}_{s(n)}^{(d)} / c_{\text{scale}}\right), \qquad L_{k,n} = \frac{1}{m}\sum_{\alpha\in\mathcal{A}} L_{k,n}^\alpha, \tag{73}$$

$$\ell_{k,n}^\alpha = L_{k,n}^\alpha / L_0, \qquad \ell_{k,n} = \frac{1}{m}\sum_{\alpha\in\mathcal{A}} \ell_{k,n}^\alpha. \tag{74}$$

The same L_0 applies to all experts and levels within a case and feedback design (Section 3.5.3.4). Score the originally computed forecasts in stateful learning and the reconstructed forecasts in a re-run.

3.5.3.2 Exponential updates and optional sharing Let $w_{k,n}^\alpha$ and λ_n^α be the weight and rate before update n. Before optional sharing, update separately at each level to obtain $\bar{w}_{k,n+1}^\alpha$:

$$\bar{w}_{k,n+1}^\alpha = \frac{w_{k,n}^\alpha \exp(-\lambda_n^\alpha \ell_{k,n}^\alpha)}{\sum_{j=1}^{K} w_{j,n}^\alpha \exp(-\lambda_n^\alpha \ell_{j,n}^\alpha)}. \tag{75}$$

With common weights, there is one update using the mean loss:

$$\bar{w}_{k,n+1} = \frac{w_{k,n} \exp(-\lambda_n \ell_{k,n})}{\sum_{j=1}^{K} w_{j,n} \exp(-\lambda_n \ell_{j,n})}. \tag{76}$$

Lower loss increases relative weight; larger λ strengthens the response. Weights start uniformly, and $\lambda = 0$ keeps equal weights.

When sharing is enabled with fraction γ, follow the exponential step by

$$w^{\alpha}_{k,n+1} = (1-\gamma)\bar{w}^{\alpha}_{k,n+1} + \gamma/K. \tag{77}$$

Omit α for common weights. Sharing helps downweighted experts recover: use $\gamma = 1/(2H)$ when enabled and zero otherwise. Table 4 distinguishes sharing from common weights across levels.

3.5.3.3 Adaptive learning rates The lookback H counts processed updates. The first H updates use starting rate λ_{start}. For rate-grid and weekly-start mixtures, record the weighted mean squared loss of the experts, using weights before the exponential step and sharing:

$$\begin{aligned} M^{\alpha}_{n} &= \sum_{k=1}^{K} w^{\alpha}_{k,n}(\ell^{\alpha}_{k,n})^2, \\ M_n &= \sum_{k=1}^{K} w_{k,n}\ell^2_{k,n}. \end{aligned} \tag{78}$$

The score mixture `score_learned` instead uses the weighted variance (`centered=TRUE`):

$$M^{\alpha}_{n} = \sum_{k=1}^{K} w^{\alpha}_{k,n}\left(\ell^{\alpha}_{k,n} - \sum_{j=1}^{K} w^{\alpha}_{j,n}\ell^{\alpha}_{j,n}\right)^2. \tag{79}$$

For this centred rule, treat variances at most $10^{-12}\sum_k w^{\alpha}_{k,n}(\ell^{\alpha}_{k,n})^2$ as zero, and analogously for common weights, to avoid large rates from numerical rounding. For per-quantile weighting, the rate is

$$\lambda^{\alpha}_{n} = \begin{cases} \lambda_{\text{start}}, & 1 \le n \le H, \\ \sqrt{\dfrac{\log(2KH)+1}{\sum_{r=n-H}^{n-1} M^{\alpha}_{r}}}, & n > H. \end{cases} \tag{80}$$

Only the previous H values of M^{α}_{r} enter; a zero sum retains the previous finite rate. Larger values reduce the rate.

Why the score mixture uses the variance. The step (75) is unchanged when the same amount is added to every expert's loss: the factor $e^{-\lambda c}$ cancels in the normalization. Only the differences between experts move the weights. The weighted mean squared loss (78), however, also grows with a loss that all experts share, and so lowers the rate (80) without any information for the weights. Since the step ignores a common shift, the losses can be centred at their weighted mean before squaring; M^{α}_{n} is then the weighted variance (79), which sets the rate by the differences alone. The score members are complete corrections of the same base, so their losses are dominated by the base error they share: in the exploratory check of the score bank the weights under the weighted mean squared loss hardly left uniform, which motivated the centred rule. In the saved runs for COVID, $h = 2$, delay 0 (the five plotted locations), the between-expert variance under equal weights is 2% of the weighted mean squared loss for the score bank, so the centred rate is about seven times larger. The same holds for the weekly starts (2%) and, less so, for the rate grid (18%); these mixtures keep the weighted mean squared loss of Gibbs and Candès [22], whose regret bound (their Lemma 2) gives the constant $\log(2KH)+1$ for $\gamma = 1/(2H)$.

Per-quantile weighting maintains a vector, a history of M^{α}_{r} and a rate for each level. Common weighting uses one of each: average losses across levels *before* computing M_r, then omit α in (80).

Later starts and implementation. For a released row s, only the weekly-start copies started at or before s are eligible: they issued a forecast for that row. Normalize their weights for the step (75), then restore their total mass; sharing (77) spreads γ times that mass over the eligible copies only. Copies started later keep their weights. Use the eligible count as K in (78)–(80). Updates with fewer than two eligible experts do not count toward H. At each reference week, first process the released feedback, then admit the new copy with $1/K$ of the enlarged bank, preserving the existing relative weights, then issue the mixture. For `restart_recent`, a copy that would make the bank larger than K_{recent} retires the oldest one first: its weight goes to the remaining copies in proportion to their weights, and it is never eligible again.

3.5.3.4 Expert loss normalization

Using the quantile loss in (1), define one reference loss per case and delay:

$$L_0 = \max\left\{\varepsilon_L, \frac{1}{20|\mathcal{A}|}\sum_{s\in I_{20}^{(d)}}\sum_{\alpha\in\mathcal{A}}\rho_\alpha\left(\frac{b_s^\alpha}{c_{\text{scale}}}, \frac{\bar{y}_s^{(d)}}{c_{\text{scale}}}\right)\right\}, \qquad \varepsilon_L = 10^{-6}. \tag{81}$$

Losses are relative to the initial mean base loss.

Initial and fixed-rate weight updates. Dividing by L_0 preserves rankings and makes weight changes depend on relative loss gaps. A change of count units scales both L and L_0 equally, leaving weights unchanged when the numerical floor is inactive. This matters throughout fixed-rate runs and during the first H adaptive-run updates at λ_{start}; these are distinct from the 20 rows used to estimate L_0.

Effect on adaptive learning rates. After H updates, the rule in (80) already adapts to the losses through M_r^α. Use superscripts raw, norm for calculations without and with L_0 normalization. Conditional on the same pre-update weight vectors at updates $n-H,\dots,n$ and a positive sum of M_r^α, normalization gives, separately at each level,

$$\begin{aligned} M_r^{\alpha,\text{norm}} &= M_r^{\alpha,\text{raw}}/L_0^2, & \lambda_n^{\alpha,\text{norm}} &= L_0\lambda_n^{\alpha,\text{raw}}, \\ \lambda_n^{\alpha,\text{norm}}\ell_{k,n}^\alpha &= \lambda_n^{\alpha,\text{raw}}L_{k,n}^\alpha. \end{aligned} \tag{82}$$

The identity holds for both definitions of M_r^α and for common weights. The numerical zero-variance threshold of the centred rule scales by L_0^{-2} too. Thus the loss and rate changes cancel in the exponential update, conditional on the same weight history. Runs with the same numerical starting rate can still differ: their first H updates change later weights and M_r^α.

Equivalent runs require matching settings and $\lambda_{\text{start}}^{\text{raw}} = \lambda_{\text{start}}^{\text{norm}}/L_0$, including the same scaling of retained rates when the sum of M_r^α is zero. Keep normalization throughout a run so incoming losses, stored M_r^α and rates remain in consistent units. After adaptation, L_0 is not an additional reduction in update strength.

3.5.3.5 Expert combinations implemented in the code

Table 4 groups the current combinations. Every learned mixture uses the adaptive rate above, with the weighted mean squared loss for rate-grid and weekly-start mixtures and the weighted variance for the score mixture.

Method IDs	Weighting and bank
`uniform`	Equal weights on 12 persistent rate experts.
`weighted`, `shared`	Same rate bank; per-quantile weights, without/with sharing.
`weighted_common`, `shared_common`	Same rate bank; common weights, without/with sharing.
`restart_uniform`	Equal weights on all weekly starts so far, rate $.05S_0$.
`restart_weighted`	Same starts; adaptive λ, sharing, per-quantile weights.
`restart_recent`	The eight most recent starts; adaptive λ, sharing, per-quantile weights; the oldest retires when a new copy starts.
`restart_recent_dual`	Two copies per start, rates $.03S_0$ and $.3S_0$; the 16 copies of the eight most recent starts; adaptive λ, sharing, per-quantile weights.
`score_equal`	Equal weights on the declared score bank.
`score_learned`	Same score bank; adaptive λ, sharing, per-quantile weights.
`power_uniform`, `tpit_uniform`, `spread_uniform`	Equal weights on the 12 rate experts of the scaled tracker, rates m in units of its scale.
`power_weighted_common`, `tpit_weighted_common`, `spread_weighted_common`	Same scaled rate bank; common weights, no sharing (as `weighted_common`).
`power_restart_equal`, `tpit_restart_equal`, `spread_restart_equal`	Equal weights on the eight most recent weekly starts of the scaled tracker, rate .05 in units of its scale.
`power_restart_recent`, `tpit_restart_recent`, `spread_restart_recent`	Same starts; adaptive λ, sharing, per-quantile weights (as `restart_recent`).

Table 4: Current expert combinations.

The four `_outer` variants of `shared`, `shared_common`, `restart_uniform`, and `restart_weighted` add persistent MultiQT at $.001S_0$ to their inner mixture, retaining its settings and learning from their own final forecasts. Rate selection and fixed $H = 8$ are described in Section 3.3.

Rate and weekly-start banks. The rate bank uses

$$\kappa \in \{0, .01, .015, .02, .03, .05, .075, .1, .15, .2, .3, .45\};$$

all experts start together and the zero-rate member is the original. The weekly-start bank adds a fresh copy of Fixed MultiQT, with rate $.05S_0$, at every retained reference week once initialization permits; with warmup replay, starts begin in the first round. The copy started at week t_k issues the base at t_k, with zero correction, and then learns only from the reports on its own forecasts, the rows $s \geq t_k$; reports on earlier rows are ignored. Older copies keep their accumulated corrections, and there is no permanent zero-rate expert. At week t the bank thus holds corrections learned over every window that ends at t and starts at a retained week; the newest copy is still the base. Starts are counted in retained reference weeks, not elapsed weeks, so a gap without forecasts adds no copies.

The three mixtures weigh this bank differently. `restart_uniform` gives all copies started so far equal weights and learns nothing; `restart_weighted` learns their weights as in (75)–(80), always with sharing $\gamma = 1/(2H)$ (unlike the rate grid, there is no weekly-start mixture without sharing). `restart_recent` learns the weights of the $K_{\text{recent}} = 8$ most recent copies $\mathcal{B}_t$ in the same way, also with sharing, or of all copies if fewer have started; older copies retire:

$$\hat{q}_t^{\alpha} = \sum_{k \in \mathcal{B}_t} w_{k,t}^{\alpha} f_{k,t}^{\alpha}, \qquad \sum_{k \in \mathcal{B}_t} w_{k,t}^{\alpha} = 1, \qquad q_t = \Pi_{\mathcal{K}}(\hat{q}_t). \tag{83}$$

This limits the age of the corrections without detecting peaks: no correction older than eight starts enters the forecast. A new copy receives weight after the released feedback is processed and before the forecast is issued, so a new color band in the weight plots alone does not show lower loss.

`restart_recent_dual` starts two copies at every retained week, a slow one with rate $.03S_0$ and a fast one with $.3S_0$. It learns the weights of the 16 copies of the eight most recent starts as in (83), with sharing; both copies of the oldest start retire together. Within the same recent window, the weights can thus choose between a slow and a fast correction.

The five score-based experts and fixed membership. The bank compares ways to retain the base or correct its errors. Write $S^{\alpha}_{k,t}(y)$ for the residual score of member k. The members are the base, the two additive trackers, the best studentized method and Delphi's square-root tracker; the base-width studentized methods are left out (Section 4.5.1). All members except `sqrt_constant` track counts; for their scores, $r = y$, projected as in (62). Fixed, leaky and studentized experts use separately selected rates, `sqrt_constant` Delphi's constant rate, and the original does not update. This compares complete procedures, not scores at a common rate.

Membership is fixed across reference weeks, horizons and re-runs, even when forecasts coincide. Do not deduplicate complete forecast histories: later forecasts or revisions must not change earlier membership.

Table 5: The declared score bank. Scales are the bounded scales defined earlier.

Member	Score or forecast	Purpose
`original`	$f^{\alpha}_{k,t} = b^{\alpha}_t$	Preserve the base as an option.
`fixed`	$S^{\alpha}_{k,t}(y) = r - b^{\alpha}_t$	Persistent additive correction; $\eta = \kappa S_0$.
`leaky`	Fixed MultiQT with leak $\delta = .1$, (61)	Additive correction that fades back to the base unless feedback renews it.
`studentized_level_rms`	$S^{\alpha}_{k,t}(y) = (r - b^{\alpha}_t)/\sigma^{\alpha}_{t,\mathrm{res}}$	Keep the full base; scale each level's correction by its recent errors.
`sqrt_constant`	$\chi(y) - \chi(b^{\alpha}_t)$, (88)	Correction on the square-root scale, between additive and level-proportional.

Initialization and scales follow Table 2. Leaky changes the state update rather than defining a new static residual score.

Equal and learned score mixtures. `score_equal` averages the declared experts. For `score_learned`, learn one vector per quantile level using the normalized losses, adaptive rate and sharing above, then repair crossings by isotonic projection.

3.5.4 Reference benchmarks

The references are the original forecast, a hindsight control and two production configurations of Delphi [23].

Original forecast: `original`. Return $q_t = b_t$ without recalibration or learning. This is the reference for the reported loss changes and reductions.

Target oracle: `oracle_target`. This hindsight control returns

$$q_t^{\text{oracle}} = \begin{cases} q_t^{\text{fixed}}, & T_t \le p(t), \\ b_t, & T_t > p(t). \end{cases} \tag{84}$$

This bypasses correction for postpeak targets, including forecasts made before the peak. Fixed MultiQT still learns from its own forecasts. The control tests bypassing correction with perfect phase information; it is not deployable. Setting the rate to zero would instead freeze the offset, giving a different control.

Delphi benchmarks: separate scales and initialization. Both run for both diseases and all feedback designs, independently of the identity/log setting. States start at zero without S_0 or warmup; constants are untuned. For transform χ and projected forecast ξ_t in those units, the released coverage-error sum is

$$\epsilon_t^\alpha = \sum_{s \in \mathcal{R}_t^{(d)}} \left(\alpha - \mathbf{1}\{\chi(\bar{y}_s^{(d)}) \le \xi_s^\alpha\}\right). \tag{85}$$

In a weekly re-run, use its current reports and reconstructed forecasts, as in Section 3.2.

Adaptive fast and slow offsets: `ref_op`. Let $\mathcal{F}_t^{(d)} = \bigcup_{j \le t} \mathcal{R}_j^{(d)}$ contain all consumed rows, and $\mathcal{F}_{t,20}^{(d)}$ the last 20 in release order, or fewer if unavailable. The COVID production configuration uses $\chi(x) = \sqrt{x}$ on raw counts, with rates

$$\begin{aligned} \eta_t^F &= \max\left\{10^{-3},\ .03Q_{.9}\left(\left\{|\sqrt{\bar{y}_s^{(d)}} - \sqrt{b_s^\alpha}| : s \in \mathcal{F}_{t,20}^{(d)},\ \alpha \in \mathcal{A}\right\}\right)\right\}, \\ \eta_t^S &= .003Q_{.9}\left(\left\{|\sqrt{\bar{y}_s^{(d)}} - \sqrt{b_s^\alpha}| : s \in \mathcal{F}_t^{(d)},\ \alpha \in \mathcal{A}\right\}\right). \end{aligned} \tag{86}$$

Both rates are zero before the first release, pool absolute base errors over levels, and are recomputed as feedback arrives. The fast and slow hidden states $\tilde{\theta}_t^F, \tilde{\theta}_t^S$ and forecast are

$$\begin{aligned} \tilde{\theta}_t^{F,\alpha} &= .9\tilde{\theta}_{t-1}^{F,\alpha} + \eta_t^F \epsilon_t^\alpha, & \tilde{\theta}_t^{S,\alpha} &= \tilde{\theta}_{t-1}^{S,\alpha} + \eta_t^S \epsilon_t^\alpha, \\ \xi_t &= \Pi_{\mathcal{K}}(\sqrt{b_t} + \tilde{\theta}_t^F + \tilde{\theta}_t^S), & q_t^\alpha &= (\max\{\xi_t^\alpha, 0\})^2. \end{aligned} \tag{87}$$

The fast correction loses 10% per processed reference week, even without feedback; gaps do not compound decay. The slow correction persists. Re-runs reconstruct both rates and states. These rolling scales are separate from the frozen S_0.

Constant square-root rate: `sqrt_constant`. The flu production benchmark uses $\chi(x) = \sqrt{x/c_{\text{pop}}}$, where c_{pop} always equals population/100,000. Its update and forecast are

$$\begin{aligned} \tilde{\theta}_t^\alpha &= \tilde{\theta}_{t-1}^\alpha + .018\epsilon_t^\alpha, \\ \xi_t &= \Pi_{\mathcal{K}}\left(\sqrt{b_t/c_{\text{pop}}} + \tilde{\theta}_t\right), & q_t^\alpha &= c_{\text{pop}}(\max\{\xi_t^\alpha, 0\})^2. \end{aligned} \tag{88}$$

The rate .018 is in square-root counts-per-100,000 units.

Delphi's adaptive rate for the other trackers. `fixed_adaptive`, `leaky_adaptive`, `fixed_log_adaptive`, `leaky_log_adaptive` and `sqrt_adaptive` replace the tuned rate of their tracker (counts, `log1p` or square root) by the fast rate η_t^F of (86), computed in the tracker's own units: .03 times the 90% quantile of the absolute base errors over the last 20 consumed rows, at least 10^{-3}. The rate rolls with the feedback, needs no warmup or hindsight and is not tuned; the leaky variants keep their leak. `sqrt_adaptive` is REF-op without its slow term and leak. There is no adaptive rate, decay or second state. Delphi selected this constant using flu seasons 2024/25 and 2025/26, which overlap our evaluation period. Although we do not retune it, this is not an independently selected held-out benchmark for those seasons.

Zero boundary and plots. Square-root methods return zero when $\xi_t^\alpha < 0$ but learn from untruncated ξ_s^α. At zero outcomes, learning coverage can differ from coverage of the returned forecast. Both appear under "Single trackers" in script 04 and in the evaluation tree of script 06.

4 Results

All results come from the run of 12 November 2026 (evening) with the methods of Section 3.5: CMU–TimeSeries submissions as the base, tuning through 31 May 2025, evaluation from June 2025 through September 2026, scored against the latest counts. Unless stated otherwise, a number is the change of a pooled mean relative to the base, in percent: of the quantile loss per 100,000 ("loss") or of the calibration error (CE). Pooled means weight every location–horizon case equally over the 10 largest states without Florida and all horizons (Section 3.1, "Locations"); negative is better. "Up to peak" and "after peak" are the windows of Section 3.4. Each feedback design uses the rates selected with it. Where a table gives one number per method and window, it is the mean over the six disease–design cells.

The tables summarize; the plots in `output/pdf/` show every disease, horizon, design and plotted location. The pages cited below:

- `multiqt_phase_rate_tradeoffs.pdf` (script 02, Fixed MultiQT on the rate grid): pp. 1–3 pooled, one page per design (delay 0, delay 1, re-run), horizons −1 to 3 in the columns, loss and calibration error of each disease in the rows; pp. 4–6 the US.
- `multiqt_error_tradeoff.pdf` (script 02): p. 1 the pooled means of every disease, design and horizon; pp. 2–7 the single locations, COVID and then flu, one page per design.
- `multiqt_tuning.pdf` (script 05): the elbow curves, pp. 1–3 COVID and 4–6 flu, one page per design.
- `multiqt_peak_reasoning_tradeoff.pdf` (script 04, all methods): per disease the 10 pooled states (pp. 1 and 8), their means only (pp. 2 and 9) and the US, California, Texas, New York and Pennsylvania (pp. 3–7 and 10–14).
- `multiqt_feedback_bars.pdf` (every method, the three designs side by side) and `multiqt_evaluation_tree.pdf` (the ten best per page), script 06: per disease the pooled page (pp. 1 and 7) and the five locations (pp. 2–6 and 8–12). The method groups of the bar plot are those of the forecast pages.
- `multiqt_method_forecasts/` (script 04), one file per disease, horizon and design: ten pages per location, one per method group ("Single trackers", "Log1p tracker", "Rate-grid experts", "Weekly starts: immediate admission", "Score-based experts", "Studentized median", "Studentized full base", "Power scale", "t-PIT scale", "Spread scale"); US pp. 1–10, California 11–20, Texas 21–30, New York 31–40, Pennsylvania 41–50.
- `multiqt_weight_timelines/` (script 04), one file per disease, horizon and design: ten pages per location in the same order (US pp. 1–10, California 11–20, and so on): rate-grid weights,

score experts, weekly starts, two-rate weekly starts, then the rate grid and the weekly starts of the power, t-PIT and spread scales.

The bar plot and the tree show the loss reduction, $100(1 - \text{pooled loss/pooled base loss})$, positive is better, with each case weighted by its weeks; the tables give the change of the mean over cases, negative is better. Overall the two agree up to the sign within 1.4 points; in the peak windows, where cases have different numbers of weeks, they mostly differ by about a point, for a few methods by up to 39 points. The score tables are in `output/results/peak_reasoning/`. Each subsection ends with the plots that show its results.

4.1 Summary

The answers in brief, as means over the six disease–design cells of the change versus the base. The brackets name the table and the plots that show each result; "not plotted" marks numbers computed only for these notes.

- **Fixed rates.** Pooled over horizons, no rate beats the base overall in any design, and the loss grows with the rate. For flu half the calibration error is gone at $\kappa = .01$; for COVID it keeps falling with the rate. Only the rise to the peak gains, and longer horizons pay more. The exception is the nowcast ($h = -1$), where $\kappa = .01$ beats the base for both diseases. [Table 6; `multiqt_phase_rate_tradeoffs.pdf`, pp. 1–3, the horizons in the columns, $h = -1$ first; `multiqt_error_tradeoff.pdf`, p. 1.]

- **Feedback designs.** Delay 0 learns from first reports, which are about 7% too low (85% of them below the latest count; not plotted). At equal rates it therefore improves calibration much less than delay 1 or the re-run (COVID at $\kappa = .01$: +16% against −25% and −37%). With the selected rates, the comparison also reflects the different rates chosen per design. [Section 4.3. Equal rates: `multiqt_phase_rate_tradeoffs.pdf`, pp. 1–3, and `multiqt_error_tradeoff.pdf`, p. 1. Selected rates: Table 7, `multiqt_tuning.pdf`, `multiqt_feedback_bars.pdf`, pp. 1 and 7.]

- **Single trackers.** Only Delphi's square-root trackers stay near the base (+0.9%, +1.6%). Fixed MultiQT costs +21% and Leaky +40%, mainly through their selected rates. The `log1p` trackers are better up to peaks and on the log scale (log loss +11% against +88% for Fixed) but lose after peaks on counts (+85% and +110%). With Delphi's untuned adaptive rate the leaky trackers lose about 2% at 10–15% less calibration error. [Tables 8 and 7; `multiqt_feedback_bars.pdf`, pp. 1 and 7, groups "Single trackers" and "Log1p tracker"; `multiqt_peak_reasoning_tradeoff.pdf`, pp. 2 and 9.]

- **Studentized methods.** None beats the base; the best, `studentized_pit`, costs +19.1%, the power scale +22.1% and `studentized_level_rms` +23.5%, against +21.4% for Fixed. The base-width scale fails after peaks (`studentized_base`, COVID +47% to +92%): the base width keeps growing after the peak, and the correction learned during the rise grows with it. [Table 9; Section 4.5.1, whose three facts are not plotted; `multiqt_feedback_bars.pdf`, groups "Studentized median" and "Studentized full base".]

- **CMU forecast versus its median.** Keeping all CMU quantiles beats rebuilding from the median by 11–36 points. Most of the value is the weekly width; the quantile shape adds 6 (flu) to 17 (COVID) points. [Table 14; in `multiqt_feedback_bars.pdf`, Fixed MultiQT and the groups "Studentized median" and "Studentized full base".]

- **Rate grid.** Learned weights beat equal weights; one weight vector for all levels without sharing is best (+6.3%), and sharing costs about 5–6 points. [Table 15; `multiqt_feedback_bars.pdf`, group "Rate-grid experts".] The weights pick the rate per location: .03–.05 in California, Texas and the US, 0 in New York and Pennsylvania; the rates .3 and .45 get little weight.

[Rate-grid weight pages, column "Learned, no sharing"; the mean weights quoted are not plotted.]

- **Weekly starts.** `restart_recent` is the best mixture (+1.9%), but its weights stay almost equal, so its gain comes from the eight-week window, not from learning. The two-rate version splits its weight about evenly between slow and fast copies: more loss (+11.1%), but the best after-peak calibration. [Table 15; `multiqt_feedback_bars.pdf`, group "Weekly starts: immediate admission"; weekly-start weight pages, column "Most recent starts: learned + sharing", and the two-rate weight pages.]
- **Score bank.** Equal and learned weights perform alike, and neither beats its member `sqrt_constant`. [Tables 15 and 8; `multiqt_feedback_bars.pdf`, groups "Score-based experts" and "Single trackers"; score-expert weight pages.]
- **Scaled methods.** With the scales of the most stationary scores, the weekly starts stay within about 1% of the base (power and spread scales) but gain less calibration than the unscaled ones (−6% to −10% against −18%); the learned rate grids reach −32% to −35% calibration error at +9% to +15% loss, and the scaled trackers alone lose 22% to 30%. The t-PIT scale is unstable. [Table 18; `multiqt_feedback_bars.pdf`, groups "Power scale", "t-PIT scale" and "Spread scale".]
- **Locations.** For COVID, California and Texas gain, and several methods beat the base there; New York, Pennsylvania and the US lose with most methods. For flu the after-peak loss decides. Calibration improves strongly for flu everywhere, but less for COVID in the large states. [Table 19; `multiqt_feedback_bars.pdf` and `multiqt_evaluation_tree.pdf`, pp. 2–6 (COVID) and 8–12 (flu), the coverage-error panels for calibration; `multiqt_peak_reasoning_tradeoff.pdf`, pp. 3–7 and 10–14.]

4.2 Fixed rates: what the trade-off plots show

Script 02 runs Fixed MultiQT, untuned, on a dense grid of rates $\eta = \kappa S_0$, $\kappa \in [.01, .45]$, for every location, horizon and feedback design. `multiqt_phase_rate_tradeoffs.pdf` shows the pooled loss and calibration error against κ by horizon, overall, up to and after peaks; `multiqt_error_tradeoff.pdf` plots calibration error against loss for each location and horizon. Table 6 averages the pooled changes over the three feedback designs.

		κ				
Disease	Change vs base (%)	.01	.03	.1	.3	.45
COVID	Loss, overall	3.5	10.4	39.2	127.2	189.5
	Loss, up to peak	−3.7	−3.9	33.9	164.7	226.1
	Loss, after peak	6.5	16.5	30.2	52.8	73.5
	CE, overall	−15.5	−29.0	−40.2	−52.3	−56.2
Flu	Loss, overall	1.1	7.4	29.9	91.5	135.4
	Loss, up to peak	−2.4	−2.9	−2.6	1.0	5.8
	Loss, after peak	1.1	5.4	24.4	78.5	105.7
	CE, overall	−50.2	−43.9	−48.2	−49.4	−52.8
	CE, after peak	4.3	12.2	29.6	32.2	23.3

Table 6: Fixed MultiQT on the rate grid (script 02), mean over the three feedback designs of the pooled change versus the base.

- **No rate beats the base overall.** In every disease and design the pooled overall loss is lowest at $\kappa = 0$ and grows with the rate: by 1–4% at $\kappa = .01$, 7–10% at .03 and 90–130% at .3.
- **Calibration: flu saturates, COVID does not.** For flu half the calibration error is gone at $\kappa = .01$, and larger rates add little while the loss keeps rising; small rates are the efficient end of the trade-off. For COVID the calibration error keeps falling with the rate (-16% at .01, -40% at .1, -56% at .45). At $\kappa = .01$ it even grows at delay 0 ($+16\%$) and at horizons 2 and 3 ($+12\%$, $+9\%$).
- **Only the rise gains.** Up to peaks, small rates lower the loss (COVID about -4%, flu about -3%); after peaks every rate raises it. For flu, the after-peak calibration error also grows with the rate up to .3: corrections learned during the rise persist into the decline.
- **Longer horizons pay more.** At $\kappa = .1$ the COVID loss rises by 11% at $h = -1$ and by 52% at $h = 3$ (flu: 9% and 33%), and the calibration gains shrink (COVID -53% against -36%, flu -60% against -26%). At the nowcast horizon, $\kappa = .01$ lowers the loss for both diseases (COVID -3.1%, flu -0.8%) and the calibration error by about half (-49%, -47%); apart from flu at $h = 0$ (-0.2%), these are the only horizons where Fixed MultiQT beats the base overall.
- **Locations** (`multiqt_error_tradeoff.pdf`) follow the same pattern: larger rates move the points towards lower calibration error and higher loss, separately up to and after peaks.

Plots. The curves behind Table 6: `multiqt_phase_rate_tradeoffs.pdf`, pp. 1–3 (pooled, one page per feedback design: loss and calibration error against κ by horizon and window; the US on pp. 4–6). The trade-off itself: `multiqt_error_tradeoff.pdf`, p. 1 (pooled means only, every disease, design and horizon) and pp. 2–7 (single locations).

4.3 Feedback designs

The base forecasts and the scoring target are the same in every design; only the reports the trackers learn from differ (Section 3.2). Two comparisons separate report timing from the selected rates.

At equal rates (script 02), delay 1 and the weekly re-run reduce calibration error far more than delay 0: at $\kappa = .01$, COVID -25% and -37% against $+16\%$, flu -55% and -51% against -45%. Up to peaks they also lower the loss more (COVID at $\kappa = .02$: -6% and -9% against -2%); after peaks they cost a little more (COVID at $\kappa = .01$: $+7\%$ and $+8\%$ against $+4\%$). The reason is the first report. Delay 0 learns from it, and it is too low: on the horizon-0 rows of the 10 states its median is 0.93 of the latest count for both diseases, 85% of the rows lie below the latest count, and 59% (COVID) and 62% (flu) more than 5% below. One week later the median is 0.98 and 68% of the rows lie below. A tracker that learns from first reports thus learns to forecast too low for the latest counts it is scored against. These report ratios are computed from the prepared reports; no plot shows them yet.

With the selected rates (script 05, Table 7), the designs also differ in the rate the elbow rule picks: for flu, Fixed MultiQT gets $\kappa = .05$ at delay 0 and in the re-run but .02 at delay 1; for COVID, .1 at both delays and .05 in the re-run. With the 10 pooled states the rule selects larger COVID rates than before (delay 0: .1 against .01 over the 52 locations), so the tuned COVID methods trade much more loss for calibration (Section 4.4). Differences between designs for tuned methods mix report timing and the selected rate.

	COVID			Flu		
Method	delay 0	delay 1	re-run	delay 0	delay 1	re-run
`fixed`	.1	.1	.05	.05	.02	.05
`leaky`	.1	.1	.1	.3	.1	.3
`fixed_log`	.02	.1	.1	.2	.1	.2
`leaky_log`	.05	.2	.1	.5	.2	.3
`studentized_level_rms`	.1	.3	.3	.3	.1	.3
`studentized_full`	.03	1*	.3	.3	.3	.3
`studentized_base`	1*	1*	.3	.3	.3	.3
`studentized_pit`	.02	.05*	.05*	.05*	.05*	.05*
`studentized_power`	.1	.1	.1	.1	.1	.1
`studentized_tpit`	.1	.003*	.03	.1	.03	.03
`studentized_spread`	.03	.3	.1	.1	.1	.1

Table 7: Selected rates (κ; for the studentized methods η in units of σ_t, for `studentized_pit` in probability units). *: edge of the grid (upper, except .003, the lower edge for `studentized_tpit`; Section 4.8).

Leaky's grid was extended from .2 to 1 after its selected rate sat at that upper edge in seven of twelve cells; with the wider grid only the `log1p` rates for flu at delay 0 move above .2 (to .3 and .5). With the leak, corrections fade, so the rule selects larger steps than for Fixed MultiQT. `studentized_pit` sits at the upper edge of its grid in five of six cells.

Plots. At equal rates: `multiqt_phase_rate_tradeoffs.pdf`, pp. 1–3, and the rows of `multiqt_error_tradeoff.pdf`, p. 1. With the selected rates: `multiqt_feedback_bars.pdf` (every method, the three designs side by side, with the tree's values and axes; pooled on pp. 1 and 7) and the first split of `multiqt_evaluation_tree.pdf` (pp. 1 and 7). The selected rates and their elbow curves: `multiqt_tuning.pdf`.

4.4 Single trackers: additive, leaky, log1p and Delphi's

Table 8 compares the single trackers; Table 7 lists their selected rates.

	Loss			Log loss	CE	Cases better (%)	
Method	overall	up to peak	after peak	overall	overall	COVID	flu
`fixed`	21.4	10.9	17.7	87.5	−42.2	7	9
`leaky`	39.9	14.6	29.0	157.7	−39.1	9	3
`fixed_log`	34.8	−8.5	85.4	11.1	−37.5	17	9
`leaky_log`	46.1	−1.9	110.4	16.4	−33.6	28	8
`ref_op`	0.9	−1.4	4.3	0.3	−17.1	37	38
`sqrt_constant`	1.6	−3.9	6.1	−0.5	−28.8	15	48
`oracle_target`	3.7	10.9	0.0	20.5	−20.5	29	61
Delphi's adaptive rate instead of the tuned one							
`fixed_adaptive`	6.2	0.0	4.9	48.9	−3.6	11	17
`leaky_adaptive`	2.1	0.8	1.8	28.2	−9.8	39	21
`fixed_log_adaptive`	8.7	−7.5	26.4	1.1	−28.2	21	17
`leaky_log_adaptive`	2.3	−2.0	8.6	−0.7	−14.7	35	27
`sqrt_adaptive`	3.1	−3.2	8.3	4.4	−17.2	13	33

Table 8: Single trackers: pooled change versus the base (%), mean over the six disease–design cells. Cases better: share of location–horizon–design cases with lower overall loss than the base.

- **Only Delphi's square-root trackers stay near the base.** `ref_op` and `sqrt_constant` change the loss by +0.9% and +1.6% and reduce calibration error by 17% and 29%. They gain up to peaks (−1%, −4%), lose a little after them (+4%, +6%) and beat the base in 15 to 48% of the cases, untuned; `sqrt_constant` is close to the base for flu (−0.2 to −0.4% in every design). Their correction scales with $\sqrt{b}$, between additive and level-proportional (Section 3.5.2).
- **Fixed MultiQT** trades +21% loss for −42% calibration error, the elbow rule's choice. Where a small rate is selected, flu at delay 1 ($\kappa = .02$), it costs only +4.7%; for COVID the rule now selects .05–.1, which costs +18% to +45% (Section 4.3).
- **Delphi's leak** costs more loss than Fixed (+40%), mainly through its larger selected rates for flu (.3 against .05 at delay 0 and in the re-run; Table 7). Where the leak should help, after COVID peaks, where both trackers have about the same rate, it loses less than Fixed in every design (+14%, +18%, +16% against +27%, +32%, +24%).
- **The log1p trackers** are better than the additive ones up to peaks (`fixed_log` −8.5% against +10.9%; flu at delay 1 −15%) and on the log scale, their own criterion (log loss +11% and +16%, against +88% and +158%). On counts, they lose after peaks (+85% and +110%; `fixed_log` for flu at delay 0 and in the re-run +150% and +178% against +10% and +11% for Fixed). This is the level scaling of Section 3.5.2: the correction is issued in proportion to the base level, which stays high after the peak while the counts fall (Section 4.5.1).
- **Knowing the peak does not rescue Fixed.** The hindsight `oracle_target`, Fixed up to each peak and the base after it, still loses 3.7%: with the selected rates, Fixed's corrections already cost loss up to peaks (+10.9%).
- **Delphi's adaptive rate** (.03 times the 90% quantile of the last 20 absolute errors on the tracker's scale, Section 3.5.4) stays much smaller than the tuned rates. The trackers then lose 2–9% instead of 21–46%, but also calibrate less: `leaky_adaptive` +2.1% loss at −10% calibration error, `leaky_log_adaptive` +2.3% at −15%, close to `ref_op`, which adds a slow term and the leak to `sqrt_adaptive`. `fixed_log_adaptive` keeps most of the calibration gain (−28%) and the gain up to peaks (−7.5%) but still loses 26% after them. None beats the base pooled; by design and disease see Table 13.

Plots. `multiqt_feedback_bars.pdf`, groups "Single trackers" and "Log1p tracker" (pooled: pp. 1 and 7); `multiqt_peak_reasoning_tradeoff.pdf`, pp. 2 and 9 (means over the 10 pooled states by design, window and horizon); the forecasts on the pages "Single trackers" and "Log1p tracker" of `multiqt_method_forecasts/` (the first two of each location's ten).

4.5 Studentized methods

	Loss			Log loss	CE	
Method	overall	up to peak	after peak	overall	overall	after peak
`studentized_level_rms`	23.5	6.2	40.6	37.9	−38.4	37.7
`studentized_full`	25.8	10.8	41.9	50.2	−40.9	11.1
`studentized_base`	37.3	24.8	56.6	73.2	−43.0	12.0
`studentized_pit`	19.1	−6.0	43.3	7.8	−31.9	19.3
`studentized`	49.1	34.2	71.5	66.1	−28.5	48.1
`studentized_zero`	52.2	38.5	75.3	65.6	−12.4	58.0
`median_constant`	57.8	38.3	40.8	145.7	−37.8	23.2
`median_constant_zero`	59.7	44.9	48.6	143.9	−34.1	40.3
`studentized_power`	22.1	3.0	37.2	57.5	−40.0	14.0
`studentized_tpit`	29.6	22.5	44.7	54.3	−37.9	6.6
`studentized_spread`	30.4	7.8	57.2	32.7	−37.8	17.3
`fixed` (reference)	21.4	10.9	17.7	87.5	−42.2	4.3

Table 9: Studentized methods: pooled change versus the base (%), mean over the six disease–design cells.

No studentized method beats the base in pooled loss. The best, `studentized_pit` (MultiQT on the PIT of the base, Table 3), costs +19.1%, less than Fixed (+21.4%), and is the only one that gains up to peaks (−6.0%); after peaks it loses 43%. Its rate sits at the upper edge of its grid in five of six cells (Table 7). `studentized_level_rms` (full base, each level scaled by its recent errors) is close to Fixed overall (+23.5% against +21.4%) and better up to peaks (+6.2% against +10.9%); its best cell is COVID at delay 0 (+8.9%). After peaks it loses more (+40.6%) and its calibration error grows (+38%), because its residual scale lags the turn. The two base-width methods lose 26% and 37%, most of it after peaks (Section 4.5.1). The methods that rebuild the forecast from the median lose 49% to 60% (Section 4.6). The three scales of the most stationary scores (power, t-PIT, spread) lose 22% to 30%; their mixtures follow in Section 4.8. The `log1p` trackers, studentized by the level, fail after peaks in the same way (Section 4.4).

Plots. `multiqt_feedback_bars.pdf`, groups "Studentized median" and "Studentized full base" (pooled: pp. 1 and 7); `multiqt_peak_reasoning_tradeoff.pdf`, pp. 2 and 9; the forecasts on the pages "Studentized median" and "Studentized full base" of `multiqt_method_forecasts/` (pages 6 and 7 of each location's ten).

4.5.1 Why the base-width scale fails after peaks

`studentized_base` uses the base-width scale (66). It reacts to the current base without feedback delay: it is large when the base is uncertain and small otherwise. It nevertheless raises the pooled loss in every disease and feedback design (overall +18% to +80%), while it reduces the calibration error about as much as `fixed`. Up to peaks it is close to the base (flu −5% to −7%, COVID re-run +2%) except for COVID at delay 0 and 1, where its selected rate is at the upper edge of the grid (+79%, +85%); after peaks it loses 40–51% for flu and 47–92% for COVID, against 3–32% for `fixed`.

Three facts explain this (medians over locations, peaks and horizons 0–3, relative to the peak week):

- The base is too low up to the peak: its median is 0.42–0.90 of the latest counts (COVID 0.69–0.90, flu 0.42–0.62), and 12–84% of the counts lie above its 0.9 quantile (COVID 20–34%). The tracker therefore accumulates an upward correction, $\tilde{u}_t > 0$.

- The base width does not turn with the counts. Two weeks after the peak the counts have fallen to 0.80 (COVID) and 0.61 (flu) of their peak value, while the width has grown to 1.18 and 1.19 of its peak-week value; four weeks before the peak it was 0.75 and 0.56.
- The forecast $b_t + \sigma_t \tilde{u}_t$ rescales the whole accumulated correction with the current scale, not only the newest step (Section 2.1.5). The upward correction learned during the rise is therefore issued 1.7 (COVID) to 2.3 (flu) times larger two weeks after the peak, when the base is already high enough (flu: median 1.06 of the counts) or too high (COVID: 1.28).

`fixed` keeps its correction constant in counts: it carries the same leftover correction past the peak, but does not enlarge it. The fast rates selected for the studentized methods (0.1–1 in units of σ_t) reinforce the effect. The scale measures the base's own uncertainty, and that lags the turn of the counts.

Standard MultiQT on the `log1p` tracker rescales its state in the same way, with the level $1 + b_t^\alpha$ in place of σ_t, (70); two weeks after the peak the base median is 1.25 (COVID) and 1.14 (flu) of its peak-week value. The results confirm it: `fixed_log` and `leaky_log` lose 85% and 110% after peaks. The square-root trackers scale by about the square root of these ratios and stay within 2% of the base's loss on average. The three facts are computed from the prepared forecasts and counts (the 10 pooled states); no plot shows the base width around peaks yet (`multiqt_base_forecast_paths.pdf` shows the base medians).

4.5.2 Which score keeps a stable α-quantile?

How the scores were selected. A tracker gets by with a small rate only if the α-quantile of its score stays put, quantile stability (Section 2.1.7). Every candidate of Table 3 was therefore judged by how much its best constant threshold drifts. For each case (disease, location, horizon) and level, the hindsight α-quantile of the realized scores $S_t^\alpha(y_t)$ over all scored weeks (latest reports, November 2024 to September 2026) is the best constant threshold. Its drift is the mean over windows of 8 consecutive weeks and over the 23 levels of $|\text{coverage} - \alpha|$, its noise the same with the weeks permuted; their ratio is 1 without drift. Coverage of a constant threshold is unchanged by any fixed increasing map of the score, so the scores differ only in how the forecast enters them (Section 2.1.6). All candidates of a case use the same weeks (a finite score for each, so at least 8 reports for the window transforms; filled rounds left out); the own-target revisions, undefined at the longest horizon, use horizons -1 to 2. The search went in three steps: the scores of the trackers in use and of the studentized methods; scales as functions of the forecast (powers of the median, the revisions, the transforms of Uniejewski with their parameter grids); then the best of them in two checks (Section 4.5.3) and in MultiQT (Section 4.5.4). One scale per idea became a method: the power $p = .5$ of the median (the level), the t-PIT (where the forecast median lies among the recent reports) and the spread per level (the forecast's own spread).

Score or scale	COVID		Flu	
	Ratio	Less drift	Ratio	Less drift
Trackers in use				
Identity, $y - b_t^\alpha$	1.67	—	1.42	—
Square root	1.62	62%	1.37	70%
`log1p`	1.62	62%	1.39	56%
Level of the forecast				
Box-Cox $\lambda = .25$ on counts	1.62	58%	1.37	60%
Box-Cox $\lambda = .75$ on counts	1.64	68%	1.39	60%
$(1 + b_t^5)^p$, $p = -1$	1.77	24%	1.49	28%
$p = .25$	1.64	70%	1.38	68%
$p = .5$ (`studentized_power`)	1.62	62%	1.37	68%
$p = .75$	1.61	60%	1.36	68%
$p = 1$	1.61	66%	1.37	64%
$p = 1.5$	1.65	56%	1.42	52%
$p = 2$	1.67	46%	1.45	56%
$(1 + b_t^\alpha)^{\cdot.75}$ per level	1.61	64%	1.37	62%
Recent errors				
Level RMS (`studentized_level_rms`)	1.64	50%	1.38	64%
Median rebuild (`studentized`)	1.64	48%	1.47	34%
Spread of the forecast				
Base width (`studentized_full`)	1.64	52%	1.31	86%
Half the base IQR	1.64	52%	1.32	78%
Spread per level (`studentized_spread`)	1.64	62%	1.32	78%
PIT of the base (`studentized_pit`)	1.63	66%	1.33	76%
Revisions of the forecast				
Revisions, every target	1.62	60%	1.35	82%
Revisions, own target ($h \leq 2$)	1.56	53%	1.30	80%
Transforms of Uniejewski (2025)				
asinh ($c = 1$), window, exact	1.62	68%	1.35	76%
asinh ($c = 1$), window, at the median	1.64	54%	1.35	76%
Box-Cox ($\lambda = 0$), window, at the median	1.64	52%	1.35	74%
Mirror log ($c = .3$), window, at the median	1.65	50%	1.37	70%
t-PIT ($\nu = 25$), window, at the median (`studentized_tpit`)	1.65	52%	1.32	86%
t-PIT ($\nu = 49$), window, at the median	1.64	50%	1.31	86%
asinh ($c = 1$), centred, exact	1.63	56%	1.32	80%
Mirror log ($c = 1/3$), centred, exact	1.64	58%	1.32	80%

Table 10: Drift of the scores of the base: ratio drift/noise, mean over the 10 largest states without Florida and horizons -1 to 3 (50 cases per disease; own-target revisions: horizons -1 to 2, 40 cases); less drift: share of cases in which the score drifts less than the identity score. 1: no drift beyond chance. `R/07_score_selection.R`.

What the table shows. No score is quantile-stable: even the best constant threshold drifts about 55 to 60% beyond chance for COVID and 30% for flu, more at longer horizons (mean over the candidates: COVID 1.34 at $h = 0$, 1.84 at $h = 3$; flu 1.28 and 1.48). The scores differ much less than they drift (COVID 1.56 to 1.77, flu 1.30 to 1.49). The reason is where the drift lies: in the location of the residual. The base forecasts too low on the rise and too high after the peak (Section 4.5.1), so the α-quantile of the residual changes sign at the peak. A positive scale or an increasing transform cannot change the sign of a residual; it only changes how much each week weighs. Scales that grow with the forecast shrink the residuals of the high weeks, where the errors are largest, and spread the misses more evenly over the season: the relative scores,

powers .5 to 1 of the median, `log1p` and the square root, drift less than the identity in 60 to 66% of the COVID cases. Scales that grow too fast ($p = 1.5$, 2) drift almost as much as no scale or more, and a scale that shrinks with the level ($p = -1$) drifts most. For flu the base's own spread is the best yardstick: the base width, half its interquartile range, the spread per level, the centred transforms, the PIT and the t-PIT drift least (1.31 to 1.33, less than the identity in 76 to 86% of the cases).

Scales as functions of the forecast. The power of the median has a shallow optimum, for COVID at $p = .75$ to 1 (1.61 against 1.67 without a scale), for flu at $p = .75$ (1.36 against 1.42); per level and as exact Box-Cox scores the same powers drift alike. The own-target revisions drift least for both diseases, but they exist only on horizons -1 to 2, where the power $p = 1$ drifts as little for COVID (1.55 against 1.56) and the base width for flu (1.29 against 1.30). The revisions of every target drift like the power scale for COVID and less for flu (1.35, less than the identity in 82% of the cases). The window transforms of Uniejewski add nothing for COVID; for flu they lie between the power and the spread scales, except the t-PIT at the median, which reacts to where the forecast median lies among the recent reports. Of the scales from recent errors, the level RMS is among the less stable, and the median rebuild is the least stable for flu. Per-level Box-Cox scales are almost never in the band (under 1% of the weeks), the centred asinh in 56% (flu 65%), the centred mirror log in 93% (95%) and the level RMS scales in 43% (44%; evaluation weeks, $h = 2$, delay 0, from the earlier search).

No optimal score yet. The best for COVID, a power .75 to 1 of the median, is not the best for flu, the forecast's spread, and none removes the drift: it is a change of sign, which no scale undoes. A simple choice for both diseases is a power of the median between .5 and .75, one scale for all levels, which keeps the band; for flu the forecast's spread drifts least. What could remove the drift is a centre that changes with the phase of the wave rather than a scale (Section 2.4, "Idea: scores with a seasonal centre").

Plots and script. `multiqt_score_selection.pdf`: p. 1 the drift ratio of every candidate with the spread over cases, p. 2 the drift against the power p, pp. 3–4 the peak-aligned coverage and pp. 5–6 the running quantile (Section 4.5.3); `multiqt_score_stationarity/`: one PDF per disease and horizon, first the pooled scores by week relative to the peak, then each plotted location over time. Tables in `output/results/score_selection/`; script `R/07_score_selection.R`. The earlier search with the paper's parameter grids, and plots of each transform's exact and derivative score over time: `agent/scripts/residual_scales.R` and `agent/scripts/score_transforms_over_time.R` (outputs in `agent/outputs/residual_scales/` and `agent/outputs/score_transforms/`).

4.5.3 Two checks of the scales: around peaks and out of sample

The drift ratio uses a hindsight threshold, counts only hits, and averages over all weeks. Two checks look at the scales from other sides. First, the coverage of the hindsight constant threshold by week relative to the nearest peak (-7 to 8), pooled over the 10 states, their peaks and horizons -1 to 3; its peak-aligned drift is the mean of $|\text{coverage} - \alpha|$ over these weeks and the 23 levels. Second, the running quantile: at each cutoff the threshold is the empirical α-quantile $\hat{Q}_t^\alpha$ of all scores reported before it, each with the report learned and the scale of its own cutoff, and the forecast inverts it,

$$q_t^\alpha = b_t^\alpha + \sigma_t \hat{Q}_t^\alpha \text{ (scale)}, \qquad q_t^\alpha = \psi_t^{-1}(\psi_t(b_t^\alpha) + \hat{Q}_t^\alpha) \text{ (transform)}, \tag{89}$$

or the base quantile function at $\hat{Q}_t^\alpha$ for the PIT. It is the forecast that quantile stability justifies (Section 2.1.7), with no rate to tune, scored like the methods (latest counts per 100,000, evaluation weeks). The base is issued before five scores are reported; crossing levels are sorted; every scale

is kept within a factor of 20 of the median of its values so far, which binds only where a scale explodes (t-PIT, when the forecast median lies far outside the recent counts). R/07 computes both checks for every candidate; Table 11 shows a selection.

	COVID				Flu			
	Overall		After peak		Overall		After peak	
Scale or score	Loss	CE	Loss	CE	Loss	CE	Loss	CE
None (identity)	9.4	84.2	3.2	0.6	5.7	12.4	7.1	8.3
$(1 + b_t^{.5})^{.5}$	7.0	51.9	5.4	2.8	3.8	5.1	15.1	28.8
$(1 + b_t^{.5})^{.75}$	6.6	38.5	6.0	2.6	7.8	−5.4	29.1	48.3
`log1p`, exact	5.5	27.8	5.4	1.2	20.5	−9.7	63.4	79.8
Base width	7.6	41.8	6.8	2.7	6.4	−5.5	24.2	21.1
Spread per level	7.8	34.2	8.5	3.2	11.3	−21.3	40.5	41.2
Level RMS	8.8	22.6	9.3	3.4	10.1	−24.4	20.8	43.9
Revisions, every target	7.8	46.2	6.1	−0.6	9.8	6.9	30.2	19.5
Revisions, own target ($h \leq 2$)	5.8	34.8	4.1	0.1	8.9	−1.3	32.5	22.3
PIT of the base	7.7	40.2	8.6	4.0	15.1	−16.2	48.7	55.4
asinh ($c = 1$), window, exact	5.7	30.2	3.1	1.1	1.1	−17.6	2.4	−2.4
Box-Cox ($\lambda = 0$), window, at the median	6.7	29.8	6.4	6.5	0.7	−18.8	2.1	−2.9
t-PIT ($\nu = 25$), window, at the median	13.5	33.7	17.9	7.0	1.1	−24.5	2.9	−1.0

Table 11: Running quantile of each scale's past scores (89): pooled change versus the base (%), the 10 largest states without Florida, horizons −1 to 3 (own-target revisions: −1 to 2), delay 0 (delay 1: similar losses). `R/07_score_selection.R`.

No static recalibration beats the base overall: every scale's quantile moves enough that a learned constant costs more than it gains, most visibly for COVID, whose base is well calibrated over the evaluation period (CE 0.066) and whose shift learned in 2024/25 does not carry over. The tracker's adaptivity is needed whatever the scale. The peak-aligned coverage shows why: for every scale and both diseases the constant threshold misses on the rise, when the counts outrun the forecast, and over-covers after the peak, when the forecast stays high. The α-quantile of the score changes sign at the peak, and a positive scale cannot change the sign of the residual: the choice of scale can make the size of the residuals comparable across levels of incidence, but it cannot remove a bias that switches from under- to overprediction. The peak-aligned drift differs little between scales (COVID 0.14 to 0.17, flu 0.12 to 0.17; lowest for the own-target revisions and, for flu, the forecast's spread, highest for the median rebuild).

Out of sample the scales part more than in the drift ratio. Scales that grow with the forecast median ($p = .75$ and 1, `log1p`, the PIT, the base width) carry the correction learned on the rise into the high forecasts just after a flu peak and lose 24 to 63% there, as in Section 4.5.1, the spread per level and the revisions 30 to 41%; $p = .5$ loses 15%. The window scales of [21], which compare the forecast median with the median and spread of the recent counts, are the most robust for flu: overall +0.7% to +1.1% loss at 18 to 25% less CE, and +2% to +3% loss after peaks. For COVID the exact `log1p` and asinh scores lose least overall (+5.5%, +5.7%, then the own-target revisions with +5.8%), the asinh also after peaks (+3.1%). A 26-week window instead of all past scores loses 7 to 21 points more for COVID (median 12) and about 5 points less for flu (up to 10), apart from the window scales, unchanged for flu. So the power scale, best by the drift ratio, is not the most robust after flu peaks, where the drift ratio, an average over all weeks of coverage only, gives the costly weeks little weight. The window-standardized asinh is the candidate that holds up in both checks for both diseases; the final comparison is MultiQT with a tuned rate for a few such scales.

Plots and script. `multiqt_score_selection.pdf`: pp. 3–4 the coverage by week relative to the peak for twelve candidates (COVID, flu), hindsight threshold and running quantile; pp. 5–6 the running quantile, loss against CE change per candidate (overall, after peaks; COVID and flu, delay 0 and 1). Tables `output/results/score_selection/running_quantile.csv`, `peak_coverage.csv` and `peak_drift.csv`; script `R/07_score_selection.R` (first computed by `agent/scripts/scale_evaluation.R`).

4.5.4 MultiQT with the finalist scales

The final check runs MultiQT itself with the finalists: no scale (Fixed MultiQT), the base width, $(1 + b_t^5)^{\cdot 5}$, the window asinh as exact transform and as derivative scale at the forecast median, and the window Box-Cox ($\lambda = 0$) at the median. One tracker serves all: standard MultiQT in the units of each round's transform, $q_t = \psi_t^{-1}(\Pi_{\mathcal{K}}(\psi_t(b_t) + \tilde{\theta}_t))$, with $\psi_t(x) = x/\sigma_t$ for a scale; for the identity it equals `multiqt()`. Each score is divided by the 90% quantile c of its absolute value over the first 20 reports, as S_0 sets the units of Fixed MultiQT, so one rate grid $\eta = m\,c$, $m = .001, \ldots, .4$, serves every scale; m is tuned by the elbow of R/05 on the tuning rows, pooled over the 10 states and horizons -1 to 3.

	COVID				Flu			
	delay 0		delay 1		delay 0		delay 1	
Scale	Loss	CE	Loss	CE	Loss	CE	Loss	CE
None (Fixed MultiQT)	35.6	−22.6	45.4	−51.1	12.1	−34.8	8.4	−45.4
Base width	4.6	5.1	50.1	−38.6	16.5	−39.7	7.1	−53.5
$(1 + b_t^5)^{\cdot 5}$	42.4	−21.8	53.1	−52.9	13.6	−38.9	14.0	−48.2
Window asinh, exact	47.7	−18.5	52.8	−49.4	5.8	−35.0	5.2	−47.2
Window asinh, at the median	62.9	−38.8	35.1	−48.0	12.0	−34.5	5.0	−44.0
Window Box-Cox, at the median	58.9	−32.9	33.1	−47.2	11.8	−36.0	5.3	−44.7
After peaks								
None (Fixed MultiQT)	26.9	−12.9	32.0	−8.8	9.8	14.9	5.2	17.1
Base width	6.7	−2.2	70.6	3.8	38.6	14.2	19.7	17.1
$(1 + b_t^5)^{\cdot 5}$	44.7	−11.4	58.1	−5.4	31.6	21.0	24.5	36.7
Window asinh, exact	46.2	−14.7	58.4	−5.6	10.6	2.4	6.7	2.5
Window asinh, at the median	53.4	−19.3	42.1	−3.1	17.8	5.4	4.7	3.9
Window Box-Cox, at the median	45.6	−19.2	37.9	−2.5	15.1	3.7	4.4	4.1

Table 12: MultiQT with the finalist scales at their tuned rates: pooled change versus the base (%), evaluation weeks, the 10 largest states without Florida, horizons -1 to 3; overall, then after peaks.

At the tuned rates the window scales lose least in three of the four cells: for flu the exact asinh (+5.8% at delay 0) and all three window scales at delay 1 (+5.0% to +5.3%), for COVID at delay 1 the window Box-Cox and asinh at the median (+33%, +35% against +45% for Fixed MultiQT); after flu peaks they keep the calibration error within +2% to +5%, where Fixed MultiQT adds 15 to 17% and the base width and the power scale lose 20 to 39%. For COVID at delay 0 the elbow picks rates of .2 to .4 for the window scales (two at the edge of the grid), and they lose 48 to 63%; the base width keeps a rate of .02 and barely moves the base.

The tuned rate decides much of this. Along the whole rate grid on the evaluation weeks (pp. 3–4 of the plots), no scale gives a better overall trade-off than Fixed MultiQT: for flu a very small rate already reaches 45 to 55% less calibration error at about 1% loss, below every tuned rate, so the tuning season again asks for too much; for COVID Fixed MultiQT is ahead overall and after peaks, and the base width is behind. After flu peaks the exact window asinh has the

best trade-off of all scales, the window scales at the median and the base width follow, and Fixed MultiQT and the power scale are worst, as the running quantile showed. So the scale matters most after flu peaks, where the window asinh is the best choice; overall the rate matters more than the scale, and a studentized method does not yet beat Fixed MultiQT with a small rate.

Plots and script. `agent/outputs/scale_multiqt/scale_multiqt.pdf`: p. 1 the tuning curves, p. 2 the evaluation at the tuned rates, pp. 3–4 every rate on the evaluation weeks (delay 0, delay 1), overall and after peaks. `agent/scripts/scale_multiqt.R`.

4.5.5 Tuning in hindsight, adaptive rates and the best rate of every scale

How well the tuning worked. Every method R/05 tunes was run with every value of its grid and scored on the tuning rows and on the evaluation weeks (delay 0 and 1, the 10 states, horizons −1 to 3). R/05's elbow on the tuning rows reproduces its saved choices in all 40 cases; the same elbow on the evaluation weeks gives the rate in hindsight. In 37 of 44 method–cell pairs the rate in hindsight is smaller than the tuned one, in one larger, in six equal. The tuned rate costs on average 11.7 (COVID, delay 0), 17.6 (COVID, delay 1), 16.6 (flu, delay 0) and 5.1 (flu, delay 1) points of loss plus calibration change against the rate in hindsight. Fixed MultiQT, for example, is tuned to $.1S_0$ for COVID at delay 0 (+35.6% loss, −22.6% CE), where $.001S_0$ would have kept the base (+0.5%, +0.2%), and to $.05S_0$ for flu at delay 0 (+12.1%, −34.8%) instead of $.01S_0$ (+0.9%, −44.8%). The tuning season, one winter of about 26 weeks, asks for faster learning than the evaluation period rewards.

Delphi's adaptive rate. The untuned `*_adaptive` trackers (Section 3.5.4) lose little, since the rolling rate stays small, and calibrate less than the tuned trackers:

	COVID				Flu			
	delay 0		delay 1		delay 0		delay 1	
Tracker	Loss	CE	Loss	CE	Loss	CE	Loss	CE
`fixed_adaptive`	9.1	60.2	7.8	14.9	4.1	−24.8	5.1	−29.5
`leaky_adaptive`	1.6	11.5	1.3	−5.2	2.9	−14.9	3.4	−18.3
`fixed_log_adaptive`	4.8	8.6	7.5	−23.4	9.0	−37.9	10.0	−40.6
`leaky_log_adaptive`	1.6	−2.4	2.3	−11.5	2.9	−15.0	2.5	−22.1
`sqrt_adaptive`	4.6	42.0	4.9	1.2	1.2	−37.7	1.3	−45.1
`ref_op`	1.1	9.4	1.3	−8.3	0.8	−21.2	0.6	−30.7
`sqrt_constant`	3.1	10.6	3.8	−27.0	−0.4	−37.2	−0.4	−39.5

Table 13: Untuned trackers, overall change versus the base (%), evaluation weeks, the 10 largest states without Florida, horizons −1 to 3.

After peaks the leaky adaptive trackers reduce the calibration error by 6 to 9% at +1% to +2% loss for both diseases; `fixed_log_adaptive` loses 33% to 38% after flu peaks. The adaptive rate thus behaves like a small tuned rate: little harm, a modest gain, and no tuning season to mislead it.

The best rate of every scale. The 66 scale options of Section 10 were run through MultiQT on the rate grid of Section 4.5.4, with the rate tuned on the tuning rows and chosen in hindsight on the evaluation weeks. With the rate in hindsight no scale beats Fixed MultiQT: it is best of all 66 options for COVID at delay 1 (+11.6% loss, −38.2% CE) and for flu at both delays (+0.9%, −44.8%; +1.2%, −54.8%), for flu followed by the power $(1 + b_t^5)^{.25}$; for COVID at delay 0 every option stays within a few percent of the base, at tiny rates. The exact window asinh,

best at the tuned rates for flu (Section 4.5.4), ranks only between 23rd and 56th in hindsight. Its advantage there was robustness to the rate: for flu its tuned result is within 3 to 5 points of its best, against 17 to 21 for Fixed MultiQT (for COVID it is not). After flu peaks it remains among the best scales in hindsight as well (+4.1% loss, −2.1% CE at delay 0, against +1.1%, +2.8% for Fixed MultiQT). So the scale changes how much a wrong rate costs and how the forecasts behave after peaks; the best achievable trade-off is set by the rate.

What the weekly starts say about the scale. The most recent weekly starts (`restart_recent`) did well, above all after peaks, because no copy remembers more than eight weeks: the mixture follows a correction whose sign changes at the peak by forgetting the old one. This is the drift no positive scale can remove (Section 4.5.3): it lies in the location of the α-quantile, not in its size. A scale should therefore be chosen for what it can do, make residuals comparable across levels of incidence, and be combined with a short memory (restarts, a leak or a small rate) for the location drift. Within a copy's eight weeks a scale tied to the forecast level rescales young offsets with the incidence, which helps on the rise, while the restart keeps the large offsets of the rise from reaching the weeks after the peak, where such scales failed. Weekly starts with a studentized tracker, for example with $(1 + b_t^5)^{.25}$ or the window asinh, combine the two and are the natural next method to test. The workflow now runs them with the power, t-PIT and spread scales (Section 4.8): with the power and spread scales they stay within 1% of the base's loss but calibrate less than the unscaled weekly starts.

Plots and scripts. `agent/outputs/tuning_hindsight/tuning_hindsight.pdf` (every grid value on the tuning rows and the evaluation weeks, tuned and hindsight choices; `agent/scripts/tuning_hindsight.R`); `agent/outputs/scale_hindsight/scale_hindsight.pdf` (all 66 options, rate in hindsight and tuned; `agent/scripts/scale_hindsight.R`).

4.6 Does the CMU forecast offer more than its median?

Three pairs of methods use the same scale and rate tuning, one correcting all CMU quantiles and one rebuilding the forecast from the base median with normal quantiles:

	Loss, overall		Full minus median (points)		
Scale: full base vs median rebuild	full	median	overall	up to peak	after peak
Constant: `fixed` vs `median_constant`	21.4	57.8	−36.4	−27.4	−23.1
Recent errors: `studentized_level_rms` vs `studentized`	23.5	49.1	−25.6	−28.0	−30.9
Base width: `studentized_full` vs `studentized_base`	25.8	37.3	−11.5	−14.0	−14.7

Table 14: Full base against median rebuild, mean over the six disease–design cells.

Yes, by a wide margin. With a constant or a recent-error scale, rebuilding from the median costs 26–36 points of loss, up to and after peaks alike. Most of this is the width: the median rebuild that keeps the CMU forecast's current width (`studentized_base`) loses 11.5 points against its full-base twin (6 for flu, 17 for COVID), and 12–20 points less than the median rebuilds with a constant or lagging residual scale. The weekly width of the CMU forecast is thus its main information beyond the median; its quantile shape adds 6 (flu) to 17 (COVID) points. Starting all levels at the median instead of at normal quantiles costs another 2–4 points of loss and 4–17 points of calibration error.

Plots. Each pair is two bars of `multiqt_feedback_bars.pdf` (Fixed MultiQT and the groups "Studentized median" and "Studentized full base"), for every design: pooled on pp. 1 and 7, by location on pp. 2–6 and 8–12.

4.7 Expert weighting

	Loss			CE	
Mixture	overall	up to peak	after peak	overall	after peak
Rate grid, 12 rates $\kappa \leq .45$					
`uniform`	24.2	12.7	20.2	−24.5	2.8
`weighted`	12.7	−3.1	19.5	−29.9	10.6
`weighted_common`	6.3	−3.6	10.9	−31.8	7.7
`shared`	17.6	13.7	26.2	−24.8	9.2
`shared_common`	12.2	9.5	18.0	−25.8	7.7
`shared_outer`	18.0	13.6	26.6	−27.1	9.5
Weekly starts					
`restart_persistent`	15.7	1.1	15.9	−38.2	9.1
`restart_uniform`	9.8	−0.2	11.4	−35.8	2.5
`restart_weighted`	11.4	2.0	13.8	−29.6	4.0
`restart_recent`	1.9	−1.5	3.3	−17.7	−7.2
`restart_recent_dual`	11.1	1.3	11.9	−17.1	−11.4
Score bank					
`score_equal`	10.9	1.4	14.7	−30.7	5.6
`score_learned`	9.3	−1.7	19.2	−24.8	0.5

Table 15: Mixtures: pooled change versus the base (%), mean over the six disease–design cells. Weights are described for the saved COVID runs at $h = 2$, delay 0 (the five plotted locations), averaged over the evaluation weeks.

Rate grid: which learning rates win. Learned weights beat equal weights clearly (`weighted` +12.7%, `uniform` +24.2%): equal weights give the fast rates as much weight as the slow ones. The best grid mixture has one weight vector for all levels and no sharing (`weighted_common`, +6.3%, −3.6% up to peaks, calibration error −32%); per-level weights add noise. Sharing costs about 5–6 points: it keeps $\gamma = 1/16$ of the mass spread over all twelve rates, a share of the equal-weight mixture. The outer MultiQT adds 0.4 points. Without sharing, the weight concentrates on two to four rates (effective number 2.2–3.9 of 12) and the rate depends on the location: around $\kappa = .03$–$.05$ in California, Texas and the US, but on the base ($\kappa = 0$: 43% and 66% of the weight) in New York and Pennsylvania. The rates .3 and .45 get almost no common weight; in the per-level weights .45 is the top expert in at most 5% of the weeks and levels, so extending the grid beyond .3 changes little.

Weekly starts: the age of the corrections. `restart_recent` is the best mixture (+1.9%; COVID +0.1% to +0.5%, flu +3% to +4%) and improves after-peak calibration for flu (−10% to −15%). Its learned weights barely move (effective number 7.7–7.8 of 8): it acts as an equal average of the eight youngest copies, so its gain comes from the age limit, not from learning. The two-rate version (`restart_recent_dual`, .03 and .3) splits its weight almost evenly between slow and fast copies (48–49% on the fast ones; effective number 15 of 16). It therefore mixes in the fast correction: more loss (+11.1%; flu +14% to +15%) but the best after-peak calibration (−11.4%; flu −16% to −21%). Over all starts, equal and learned weights behave alike (+9.8%, +11.4%); the learned weight spreads over copies of every age, 30–32% on copies older than 32

weeks. The first start alone (`restart_persistent`) loses most (+15.7%). The weights of the weekly starts barely move because their losses share a large common part: the between-copy variance is about 2% of the weighted mean squared loss that sets the rate (Section 3.5.3.3); the weighted variance of the score mixture was not tried for them.

Score bank. Equal and learned weights perform alike (+10.9%, +9.3%); learned weights calibrate less (−25% against −31%). The base carries the largest learned weight (0.27–0.36), the four trackers 0.11–0.27 each, with about two members dominating any week and level (effective number 1.9–2.0 of 5). Neither mixture beats its own member `sqrt_constant` (+1.6%).

Mixtures against a single fixed rate. Tables 16 (overall) and 17 (after peaks) compare two mixtures with Fixed MultiQT on the rate grid of script 02, in each disease–design cell, pooled over the 10 largest states without Florida (recomputed with the current code and tuned rates). A fixed rate beats a mixture if both its loss and its calibration error are at most the mixture's.

		`weighted_common`			`restart_recent`		
Disease	Design	Loss	CE	Beaten by: κ (loss, CE)	Loss	CE	Beaten by: κ (loss, CE)
COVID	delay 0	6.1	−0.2	—	0.5	−7.1	—
	delay 1	8.0	−33.5	—	0.1	−19.8	—
	re-run	8.3	−37.6	.015 (5.0, −41.6)	0.2	−16.9	—
Flu	delay 0	3.9	−31.0	.01 (0.9, −44.8)	3.9	−13.7	.01 (0.9, −44.8)
	delay 1	1.8	−42.8	.01 (1.2, −54.8)	3.2	−25.5	.01 (1.2, −54.8)
	re-run	10.0	−45.9	.01 (1.1, −51.0)	3.5	−23.0	.01 (1.1, −51.0)

Table 16: Best rate-grid mixture and most recent weekly starts against single fixed rates: pooled overall change versus the base (%), the 10 largest states without Florida. Beaten by: the rate $\kappa > 0$ of the grid with both loss and CE at most the mixture's (the lowest loss if several); —: no such rate.

The best rate-grid mixture, `weighted_common`, is beaten by a single rate of .01 or .015 in four of the six cells: every flu cell and COVID in the re-run. For COVID at delay 0 it is ahead: a fixed rate needs +9.8% loss ($\kappa = .03$) for its calibration error; at delay 1, $\kappa = .02$ comes within 0.1 points of its calibration error at less loss (+7.3% against +8.0%). The other rate-grid mixtures (`uniform`, `weighted`, `shared`, `shared_common`) are beaten in every cell except `weighted` for COVID at delay 0, the score mixtures in all but COVID at delay 0. The beating rate is picked in hindsight, but $\kappa = .015$ throughout would have beaten the mixture on the means (+4.0% and −34.2% against +6.3% and −31.8%). Mixing rates thus protects against a badly selected rate (`fixed`, +21.4%), but does not improve on a small fixed one.

		`weighted_common`			`restart_recent`		
Disease	Design	Loss	CE	Beaten by: κ (loss, CE)	Loss	CE	Beaten by: κ (loss, CE)
COVID	delay 0	11.9	−1.4	.01 (4.4, −3.7)	2.4	−4.6	—
	delay 1	15.9	7.5	.01 (7.1, 4.1)	2.5	0.2	—
	re-run	16.8	7.9	.01 (7.9, 7.0)	2.8	−2.6	—
Flu	delay 0	5.3	8.8	.01 (1.1, 2.8)	4.7	−11.2	—
	delay 1	2.3	7.0	.01 (0.9, 4.9)	3.1	−10.2	—
	re-run	13.5	16.6	.01 (1.4, 5.1)	4.5	−15.1	—

Table 17: As Table 16, after peaks (weeks 1 to 8).

After peaks (Table 17), the smallest rate, $\kappa = .01$, beats the rate mixture in every cell, and the base itself does in five: the rates above .01 that the mixture weighs carry the correction learned during the rise past the peak (Section 4.2). `restart_recent` is beaten by no rate after peaks. For COVID it loses less than every rate (+2.4% to +2.8%) and changes the calibration error little (−4.6% to +0.2%); for flu it reduces the calibration error by 10 to 15% at +3.1% to +4.7% loss, while every fixed rate increases it there (+2.8% or more). Overall, no rate beats it for COVID either: its loss (+0.1% to +0.5%) is below that of every rate, at 7 to 20% less calibration error than the base. For flu, $\kappa = .01$ beats it overall in every design: it trades calibration over the season for calibration after peaks. Its weights stay almost uniform, so these gains come from the eight-week memory of its copies, not from learning which copy to trust. Of the single trackers, neither REF-op nor `sqrt_constant` is beaten by a fixed rate overall in any cell.

How much these comparisons depend on the weeks scored. The tables compare methods on one realized sequence (Section 1). A circular block bootstrap over the 69 target weeks of the evaluation period recomputes the pooled changes from the issued weekly scores: blocks of 8 weeks, one draw for all locations, horizons and both diseases, 2000 resamples; the trackers are not re-run. Calibration error averages $|\text{coverage} - \alpha|$, so resampling inflates it most for the best-calibrated methods; the draws are centred on the observed changes to remove this bias. Below, "dominates" means both loss and CE at most the mixture's, and intervals are 90% intervals of the differences fixed rate minus mixture, in percentage points of the base.

- `weighted_common`, overall: in the four cells where a rate beats it, that rate dominates it in 68 to 79% of the resamples, some rate of the grid in 75 to 96%; for COVID at delay 0 some rate does in 12%, at delay 1 in 50%. The rate's lower loss is fairly stable (intervals ending between 0.0 and +1.0), its calibration is not (CE intervals 19 to 33 points wide).
- `weighted_common`, after peaks: $\kappa = .01$ has the lower loss in every cell, with intervals below zero in four of six (flu, delay 0 and 1: up to +0.3 and +0.4), and dominates in 46 to 92%.
- `restart_recent`: after peaks, $\kappa = .01$ dominates it in at most 8% of the resamples; its flu CE advantage of 14 to 20 points has intervals that reach about zero (lower ends −2.0 to +0.3). Overall, $\kappa = .01$ dominates it for COVID in 0 to 9% and for flu in 85 to 94%.
- $\kappa = .015$ throughout against `weighted_common`, mean of the six cells: loss -2.4 $[-5.7, 0.2]$, CE -2.4 $[-8.0, 4.3]$, dominating in 73%: the lower loss is likely, the better calibration less so.

Blocks of 4 or 13 weeks give similar shares, except flu: 13 weeks lower them for `weighted_common` overall to .38–.50 and after peaks at delay 1 to .18. All flu peaks lie in one winter: its after-peak weeks span 3 January to 11 April 2026, and 3% of the resamples contain none (dropped). Resampling the 10 states instead, with all weeks kept, gives much narrower intervals that understate the uncertainty, since the locations share their epidemic weeks. What holds up: after peaks a small fixed rate beats the rate mixture on loss, and `restart_recent` is beaten by no rate; overall, $\kappa = .01$ beats `restart_recent` for flu but not for COVID. That a rate of .01 or .015 beats `weighted_common` overall in four cells is the more likely outcome there but not established: its loss advantage is fairly stable, its calibration advantage within the noise of 69 weeks. Script: `agent/scripts/mixture_vs_fixed_bootstrap.R`.

Plots. Performance: `multiqt_feedback_bars.pdf`, groups "Rate-grid experts", "Weekly starts: immediate admission" and "Score-based experts" (pooled: pp. 1 and 7). Weights: `multiqt_weight_timelines/`, per disease, horizon and design: the rate grid on pp. 1, 11, 21, 31 and 41 (columns "Equal weights", "Learned, no sharing" and "Learned + sharing"; the rows "All levels" show their versions with common weights), the score experts on pp. 2,

12, 22, 32 and 42, the weekly starts on pp. 3, 13, 23, 33 and 43 (`restart_recent` in the column "Most recent starts: learned + sharing") and the two-rate weekly starts on pp. 4, 14, 24, 34 and 44. Their forecasts: the pages "Rate-grid experts", "Weekly starts: immediate admission" and "Score-based experts" of `multiqt_method_forecasts/` (the third to fifth of each location's ten). The effective numbers and mean weights quoted above summarize the weights for COVID, $h = 2$, delay 0 (`covid_horizon2_delay0.pdf`) over the evaluation weeks; they are not plotted (`agent/scripts/weight_stats_2026-11-12.R`). Neither are Tables 16 and 17 (`agent/outputs/mixture_vs_fixed_bootstrap/tables.csv`); their mixture columns equal the run's pooled numbers, and the fixed rates are the curves of `multiqt_phase_rate_tradeoffs.pdf`, pp. 1–3.

4.8 Scaled methods: power, t-PIT and spread scales

The three scales of the most stationary scores (67)–(69) each drive a tracker with its tuned rate (Table 7) and four mixtures: the rate grid with equal and with learned common weights, and the eight most recent weekly starts with equal and with learned weights (Table 4).

	Loss			Log loss	CE		Better (%)	
Variant	overall	up to peak	after peak	overall	overall	after peak	COVID	flu
Power scale								
Tracker	22.1	3.0	37.2	57.5	−40.0	14.0	7	11
Rate grid, equal	15.5	3.4	28.6	41.6	−29.8	5.8	15	19
Rate grid, learned common	10.5	−7.8	25.8	12.8	−35.1	14.1	13	18
Weekly starts, equal	0.8	−2.2	4.3	2.2	−10.0	−6.0	53	37
Weekly starts, learned	0.9	−2.1	4.3	2.4	−10.1	−6.3	52	35
t-PIT scale								
Tracker	29.6	22.5	44.7	54.3	−37.9	6.6	17	19
Rate grid, equal	55.0	56.3	73.7	97.5	−29.5	−1.2	9	12
Rate grid, learned common	8.8	−1.7	17.2	9.4	−33.7	4.3	9	31
Weekly starts, equal	5.0	1.4	10.3	11.4	−14.7	−5.6	37	41
Weekly starts, learned	4.9	1.3	10.2	11.6	−14.7	−6.0	37	41
Spread scale								
Tracker	30.4	7.8	57.2	32.7	−37.8	17.3	4	11
Rate grid, equal	18.0	−0.6	40.5	18.7	−31.0	7.0	6	19
Rate grid, learned common	15.4	−8.5	38.1	7.9	−32.4	14.9	6	18
Weekly starts, equal	1.0	−1.0	4.4	−0.5	−5.8	−4.8	51	36
Weekly starts, learned	1.0	−1.0	4.5	−0.5	−5.8	−5.3	51	36
Unscaled, for reference								
`fixed`	21.4	10.9	17.7	87.5	−42.2	4.3	7	9
`weighted_common`	6.3	−3.6	10.9	22.4	−31.8	7.7	9	14
`restart_recent`	1.9	−1.5	3.3	22.5	−17.7	−7.2	53	25

Table 18: Scaled methods: pooled change versus the base (%), mean over the six disease–design cells. Tracker: `studentized_<scale>`; rate grid: `<scale>_uniform`, `<scale>_weighted_common`; weekly starts: `<scale>_restart_equal`, `<scale>_restart_recent`, with `<scale>` one of `power`, `tpit`, `spread`. Better: cases better, as in Table 8.

- **Weekly starts stay at the base and calibrate less.** With the power and spread scales the most recent weekly starts change the loss by +0.8% to +1.0%, less than the unscaled `restart_recent` (+1.9%); with the power scale they beat the base for COVID in every design (−0.1% to −0.4%) and in about half the COVID cases. They reduce the calibration error less

(−6% to −10% against −18%) and lose a little more after peaks (+4.3% against +3.3%); on the log scale they are much better (+2% and −0.5% against +22.5%). The scales thus move the weekly starts towards the base along the trade-off, not past it. Equal and learned weights give the same result: the learned weights stay uniform (effective number 7.6–8.0 of 8).

- **The learned rate grids gain up to peaks and lose after them.** `power_weighted_common` and `spread_weighted_common` gain 8% up to peaks, against 4% unscaled, but lose 26% and 38% after them, against 11%: the scales grow with the forecast and, like the base width (Section 4.5.1), issue the correction learned in the rise larger after the peak. Overall they lose 10.5% and 15.4% against 6.3%, at slightly more (−35%, power) or the same calibration gain. A single fixed rate beats the learned grid in five of six cells (power), four (t-PIT) and six (spread); the scaled weekly starts are beaten only in the flu cells (power, spread, by $\kappa = .01$) or in all but COVID at delay 0 (t-PIT). Equal weights lose more, the trackers alone most (22% to 30%, with the tuned rates).

- **Which scale.** The power scale is the best of the three on counts: the best weekly starts and the most calibration gain of the learned grids. The spread scale is best on the log scale (weekly starts −0.5%, learned grid +7.9%) and worst after peaks on counts. The t-PIT scale is unstable (below); its learned grid (+8.8%) and weekly starts (+5%) survive it, presumably because the learned weights move to $\kappa = 0$ and the weekly starts keep only eight weeks of corrections.

- **Known issue: the t-PIT scale explodes.** The scale $1/\hat{F}'_t(b_t^{.5})$ becomes astronomically large (raw values up to 10^{36}) when the forecast median lies outside the range of the last 26 reports, where the kernel density is almost zero: in the late-summer 2026 trough, when the counts fall below every forecast median, and for California COVID at delay 1 in March 2025. The guard of `normalized_scale()` keeps the scale within 20 times the median of its values so far, but that median includes the current week and can itself be inflated early on. In California (COVID, $h = 3$, delay 1) the explosion falls in the S_0 rows: the reference median becomes 3.7×10^7, so the normalized scale is about 0 in most weeks and up to $10^{13} S_0$ in three. The pooled tuning loss for COVID at delay 1 is then about 6×10^8% at every rate, and R/05 picks .003 at the lower edge of the grid (Table 7). Elsewhere the guard caps the scale at 20 to 36 times S_0 in the late-summer weeks (California and Texas, COVID $h = 3$), still large enough to blow up the fast rates: `tpit_uniform` loses 85%, 127% and 74% for COVID, and `studentized_tpit` 109% at delay 0. The learned grid escapes by moving its weight to $\kappa = 0$ (California, COVID $h = 2$, delay 0: 0.86). The results of the t-PIT scale therefore reflect this implementation more than the score; see the open issue in Section 2.4.

Plots. `multiqt_feedback_bars.pdf`, groups "Power scale", "t-PIT scale" and "Spread scale" (pooled: pp. 1 and 7); the forecasts on the pages of these groups in `multiqt_method_forecasts/` (pages 8 to 10 of each location's ten); the weights in `multiqt_weight_timelines/`, pages 5 to 10 of each location's ten (rate grid and weekly starts per scale). The weight statistics and the diagnosis of the t-PIT scale are not plotted (`agent/scripts/weight_stats_2026-11-12.R`, `agent/scripts/tpit_explosion_check.R`).

4.9 Locations

Disease	Location	fixed	fixed _log	sqrt _const.	ref _op	stud. level_rms	weighted _common	restart _recent	restart _dual	score _equal
COVID	10 pooled	32.8	16.6	3.6	1.1	24.6	7.5	0.3	7.7	11.1
	US	37.6	19.9	4.3	1.6	22.3	15.1	−1.4	5.8	10.4
	California	18.0	9.6	2.4	−4.5	28.4	9.9	−8.7	−11.4	2.9
	Texas	24.7	20.1	4.8	−2.1	35.5	10.7	−5.9	−6.7	7.7
	New York	37.3	6.7	1.9	−0.4	10.3	3.9	1.2	13.5	6.8
	Pennsylvania	34.3	18.0	4.2	3.4	21.4	6.0	2.6	11.8	13.2
Flu	10 pooled	9.9	53.0	−0.3	0.7	22.3	5.2	3.6	14.5	10.7
	US	9.6	60.3	−1.4	−1.2	18.8	15.4	1.7	10.4	7.2
	California	7.8	23.5	−7.9	−6.6	9.9	14.1	−1.8	5.3	0.6
	Texas	7.5	24.1	0.6	0.6	4.8	2.4	2.0	9.8	5.7
	New York	6.3	28.1	−2.7	−0.8	29.2	2.1	0.0	2.9	9.6
	Pennsylvania	11.8	61.8	−1.7	0.4	29.2	4.6	6.1	21.9	13.7

Table 19: Overall loss change versus the base (%) by location, mean over the three feedback designs (pooled: the 10 largest states without Florida).

- **COVID splits the plotted locations.** California and Texas gain most up to peaks (Fixed −35% and −8%, `restart_recent` −16% and −15%), and several methods beat the base overall there (`restart_recent` −8.7% and −5.9%, the two-rate starts −11.4% and −6.7%, REF-op −4.5% and −2.1%); they also lose most after peaks (Fixed +55% and +37%). New York, Pennsylvania and the US lose up to peaks as well (Fixed +29% to +67%); there only the Delphi trackers and `restart_recent` stay within about 4% of the base.
- **Flu** gains up to peaks in all plotted locations, least in Texas (Fixed −0.4%), and the after-peak losses decide: the `log1p` trackers lose most (after peaks +177% in the US, +157% in Pennsylvania), while `sqrt_constant` and REF-op beat the base in California (−7.9%, −6.6%), New York and the US.
- **Calibration** improves strongly for flu everywhere (Fixed −40% to −63% in the plotted locations). For COVID the large plotted locations gain less (Fixed −15% to −25%, pooled −40%), and REF-op even worsens their calibration error (+7% to +19% in the US, California, Texas and Pennsylvania).
- Pooled results thus hide large differences: for COVID the ranking of methods in California and Texas is not the one in New York and Pennsylvania. The US, as the sum of the states, resembles New York and Pennsylvania for COVID.

Plots. `multiqt_feedback_bars.pdf`, pp. 2–6 (COVID) and 8–12 (flu), every method per location; the location pages of `multiqt_evaluation_tree.pdf` (pp. 2–6 and 8–12) and of `multiqt_peak_reasoning_tradeoff.pdf` (pp. 3–7 and 10–14); the fixed rates per location in `multiqt_error_tradeoff.pdf`, pp. 2–7.

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